Band Sampling for Hyperspectral Imagery

Chein-I Chang, *Life Fellow, IEEE*

***Abstract*— Band sampling (BSam) is an innovative concept for hyperspectral imaging, which is derived from signal sampling in communications/signal processing as well as sampling theory in information theory. It is quite different from band selection (BSel) in several aspects. First of all, BSam "samples" bands with a given fixed BSam rate (BSamR) as opposed to BSel, which "selects" appropriate bands according to the number of bands to be selected. Second, BSam requires no specific means of sampling bands compared to BSel, which requires a specific rule to select bands such as band prioritization (BP) criteria or band search strategies. Third, BSam samples bands without prior band knowledge or specific applications in contrast to BSel, which considers certain bands more significant than other bands according to various applications. Two strategies are developed for BSam. One is from information theory. Under a completely blind and unknown environment, the maximum entropy is achieved by the uniform distribution. This suggests that one best strategy for BSam is uniform band sampling (UBSam). Another strategy is random BSam (RBSam) analogous to random signal sampling in compressive sensing (CS) where the bands sampled by RBSam are random but are not deterministic like the bands selected by BSel. Interestingly, UBSam and RBSam generally perform better than custom-designed BSel methods. To illustrate its potential utility of BSam in various applications, target detection, anomaly detection, and image classification are studied through extensive experiments for demonstration.**

***Index Terms*— Band sampling (BSam), iterative fixed uniform band sampling (FUBSam), iterative random band sampling (IRBSam), iterative uniform band sampling (UBSam), random band sampling (RBSam).**

NOMENCLATURE

BS	BKG suppressibility
BSam	Band sampling.
BSamR	Band sampling rate.
BSel	Band selection.
CEM	Constrained energy minimization.
CRD	Collaborative representation-based anomaly detector.
CS	Compressive sensing/sampling.
CSBD	Compressively sensed band domain
EPF	Edge preserving filter.
FTS	Fixed training sampling.
IBSam	Iterative band sampling.
EUBSam	Iterative exhaustive uniform band sampling.
FUBSam	Fixed uniform band sampling spectrometer.
IEPF	Iterative edge preserving filter.
IFBSam	Iterative fixed band sampling.
IRBSam	Iterative random band sampling.
IRS	Iterative random sampling.
ITS	Iterative training sampling.
RBSam	Random band sampling.
RTS	Random training sampling
RUBSam	Iterative random uniform band sampling.
RXD	Reed–Xiaoli detector.
SF	Spatial filter.
SFMaps	Spectral-filtered maps.
UBSam	Uniform band sampling.
UBSel	Uniform band selection.

Manuscript received March 1, 2021; revised April 24, 2021, June 28, 2021, and July 31, 2021; accepted August 3, 2021. Date of publication August 17, 2021; date of current version January 26, 2022. This work was supported by the Fundamental Research Funds for the Central Universities under Grant 3132019341.

The author is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland at Baltimore, Baltimore, MD 21250 USA, and also with the Department of Computer Science and Information Management, Providence University, Taichung 02912, Taiwan (e-mail: cchang@umbc.edu).

Digital Object Identifier 10.1109/TGRS.2021.3102861

I. INTRODUCTION

WITH utilizing hundreds of contiguous spectral bands, a hyperspectral image generally contains rich abundant but also enormous redundant spectral information. BSel has been shown to be an effective means of reducing data volumes while retaining crucial unaltered spectral information [1]. It has been widely used to reduce data dimensionality by removing redundant or unwanted bands while maintaining a certain desired level of performance. Many various approaches have been developed and reported in the literature. Basically, BSel can be categorized according to strategies of how to select bands. A simple and also effective approach is band prioritization (BP) that ranks all bands according to certain criteria, such as variance, signal-to-noise ratio entropy, information divergence criteria [2], and constrained BSel [3]. A second approach is band clustering based on grouping bands with similar spectral properties into clusters, and then, one representative band is selected from each cluster such as [4]–[6]. A third approach is to adopt certain band search strategies to find optimal bands in a feedforward or backward manner, such as minimum estimated abundance covariance (MEAC) [7], orthogonal projection-based BSel [8], and constrained target BSel [9]. A fourth approach is band subset selection (BSS), also referred to as band combination, which selects an optimal band subset of multiple bands as a whole instead of finding bands one at a time, such as particle swarm optimization [10], firefly algorithm [11], dominant set extraction band selection (DSEBS) [12], multigraph determinantal point process (MDPP) [13], channel capacity-based BSS [14], band subset for anomaly detection [15], linearly constrained

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

minimum variance-based BSS for classification [16], and optimal neighborhood reconstruction (ONR)-based BSS based on correlated neighborhood property (CNP) to exploit strong correlation between the neighboring bands [17].

Two major key issues arising in all the above-mentioned approaches are requirement of prior knowledge of how many bands needed to be selected, i.e., number of bands to be selected n_{BS}, and how to take care of interband correlation among selected bands. Regarding the first issue, finding an appropriate n_{BS} still remains a challenging task. As for the second issue, BSS has been developed for this purpose. Unfortunately, its selected band subsets or combinations are very limited due to infeasibility of exhausting all possible band combinations as band subsets.

This article looks into an interesting notion, called opposed to BSel for distinction. Its concept has never been explored in the hyperspectral imaging literature. It deviates from the conventional BSel in terms of how bands are being "sampled" by BSam not "selected" by BSel. It offers a relatively simple but yet an effective means of finding desired bands.

In analogy with BSel, BSam also has two similar issues. One is BSamR, defined by the ratio of number of bands to be sampled n_{BSam} to the total number of spectral bands L as

$$\mathrm{BsamR} = \frac{n_{\mathrm{BSam}}}{L} \tag{1}$$

which is similar to the Nyquist rate. The definition of (1) can also be referred to as the information rate described by Shannon's first noiseless source coding theorem in information theory [18]. A second one is how to determine n_{BSam}. Fortunately, the virtual dimensionality (VD) developed in [19], [20] can be used to estimate n_{BSam}. However, since VD only provides an estimate of n_{BSam}, in reality, it may be underestimated in practical applications. In this case, an issue similar to the issue of sampling rate lower than the Nyquist rate may occur. In the context of signal sampling, this issue is known as a signal aliasing problem. Recently, this issue has been resolved by random signal sampling in CS [21]–[24]. However, for CS to accomplish this task, it must satisfy two constraints: signal representations to be sparse and sampled signals to be incoherent. According to [23], if there is no concern on signal recovery, the signal sparse representation by a sparse matrix $\mathbf{\Psi}$ can be implicitly included in the so-called universality model, which only involves the sensing matrix $\mathbf{\Phi}$ with no need of referring to the sparse matrix $\mathbf{\Psi}$. This universality model greatly reduces the complexity of BSam by turning this issue to only focusing on how to sample bands to achieve the maximal signal incoherence.

In order to make BSam work, two strategies are explored in this article. It is known from information theory [18] that maximal entropy is achieved by the uniform distribution. In translation to the BSam language, it means that for given a n_{BSam}, the maximal incoherence can be achieved by sampling bands uniformly. This leads to an approach to BSam, called UBSam. However, if n_{BSam} is undersampled (viz., n_{BSam} is smaller than it should be in reality) like signal sampling rate lower than the Nyquist rate, UBSam may not work effectively. In this case, we need to appeal for CS, which uses random signal sampling to deal with the undersignal sampling rate. The concept of random sampling in CS gives rise to a second approach, called RBSam. Unfortunately, a direct use of sensing matrix $\mathbf{\Phi}$ in CS via the universality model cannot be applicable to RBSam. We must reinvent the wheel by coming up with a new idea to tackle this problem. Interestingly, a recent work in [25] developed ITS that indeed provides a possible solution. In other words, the idea of using iterative sampling of training samples can be readily applied to random sampling of bands if we interpret training samples as bands. The idea of ITS is to use a set of SFs to capture spatial classification information produced by a classifier and then feed the spatial filtered classification maps back to be added to the current data cube to create a new expanded data cube. ITS is then once again implemented on this new data cube to reselect random training samples that contain more spatial classification information via feedback of spatial filtered classification maps to reduce misclassification error. The same process is repeated iteratively. Since BSam is a band processing and does not take into the spatial information account, the way that ITS introduces spatial information may be in fact applicable to BSam if training samples are replaced by sampled bands. By taking advantage of ITS, BSam becomes realizable to correspond to its counterpart, random sampling in CS.

There are several significant contributions derived from BSam.

1) BSam samples bands according to a given fixed n_{BSam} specified by BSamR in (1) in contrast to BSel, which uses n_{BS} to select appropriate bands. By virtue of (1), we can also define a corresponding concept, BSel rate (BSelR) by $\mathrm{BSelR} = (n_{\mathrm{BSel}}/L)$ for BSel. If n_{BSel} is too small and less than what it should be, BSel methods may not work effectively no matter how good BSel methods are designed. This inherent issue has been overlooked and not been investigated in the past. However, unlike BSel, RBSam can still make BSam work even if n_{BSam} runs into the same issue.
2) BSam does not require any specific means of sampling bands compared to BSel, which requires a specific rule such as BP criteria or BSel/search strategies.
3) One key difference of BSam from BSel is that bands sampled by BSam are random and cannot be reproduced as opposed to bands selected by BSel which are deterministic and can be reproduced. The beauty of random signal sampling is derived from CS, which allows signals to be sampled signals under the Nyquist rate with two constraints, sparsity and incoherence, but does not affect the end results. Thus, including ITS in BSam also allows BSam to perform effectively in the same way that random signal sampling does in CS.
4) BSam samples each band equally likely to maximize band incoherence (i.e., interband decorrelation), whereas BSel considers certain bands that are more significant than other bands. Consequently, UBSam is used in the same manner that the uniform signal sampling is used by the Nyquist rate. This also explains why the UBSel generally performs well compared to many BSel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

methods as long as n_{BSel} is appropriately determined and greater than the true value of n_{BSel}.

5) Two key approaches are developed for BSam.
   a) One is UBSam to be discussed in Section IV, which can be further refined to RUBSam, FUBSam, and EUBSam. In particular, UBSel can be shown to be a special case of FUBSam.
   b) The other is IBSam to be discussed in Sections V and VI, which includes IRBSam and IFBSam. Interestingly, any BSel method can also be shown to be a special case of fixed BSam (FBSam) without implementing an iterative process.
6) The close relationship of BSam with VD is discussed in great detail in Section VII. In particular, VD can be used to estimate n_{BSam}.
7) BSam has a great generalizability to be discussed in Section VIII, which extends BSel to a next level of band processing.

The remainder of this article is organized as follows. Section II briefly reviews related works. Section III describes the major concept of BSam. Section IV–VI derive UBSam, IRSam, and FBSam, respectively. Section VII studies the complexity analysis of BSam and its relationship with VD, which bridges the two concepts of VD and BSam. Section VIII discusses the generalizability of BSam. Sections IX–X present the applications to target/anomaly detection and hyperspectral image classification respectively. Section XI provides the examples to illustrate the usefulness of SFs in BSam. Finally, Section XII summarizes the benefits of BSam and its potential in future directions.

II. Related Works

A new concept generally has its root and does not come from nowhere. The key idea of BSam is derived from signal sampling with RBSam corresponding to random signal sampling arising in compressive sampling/sensing [21], [22].

A. Brief Review of Compressive Sampling

According to conventional signal sampling, Shannon's sampling rate, also called Nyquist rate, provides a theoretical limit on analog-to-digital conversion. If the sampling rate is under the Nyquist rate, the sample signals cannot completely reconstruct the original signal without aliasing. To achieve this goal, two principles are imposed on CS, sparsity, which expresses a signal in a sparse representation and incoherence, which senses or samples signals as incoherently as possible.

To take care of the sparsity and incoherence, we need to jointly consider $\mathbf{\Phi}$ and $\mathbf{\Psi}$ as a pair of $(\mathbf{\Phi}, \mathbf{\Psi})$, where $\mathbf{\Phi}$ is used for sensing signals and $\mathbf{\Psi}$ is used to represent signals. In other words, $\mathbf{\Psi}$ represents a signal in a sparse representation, while the coherence measures the largest correlation between any two elements of $\mathbf{\Phi}$ and $\mathbf{\Psi}$. If $\mathbf{\Phi}$ and $\mathbf{\Psi}$ contain correlated elements, the coherence is large. Otherwise, it is small.

Based on the CS theory, the lesser the coherence between $\mathbf{\Phi}$ and $\mathbf{\Psi}$ is, the better signal sensing and reconstruction are [21], [22]. Mathematically, their mutual coherence is defined as

$$1 \le \mu(\mathbf{\Phi}, \mathbf{\Psi}) = \sqrt{N} \max_{1 \le i, j \le N} \left| \langle \phi_i, \psi_j \rangle \right| \le \sqrt{N} \tag{2}$$

where N is the signal dimensionality referred to as spectral dimensionality L in this paper and $\boldsymbol{\phi}_i$ and $\boldsymbol{\psi}_j$ represent the columns of $\mathbf{\Phi}$ and $\mathbf{\Psi}$, respectively. Specifically, let the parameter m be the measurements used in $\mathbf{\Phi}$. Then, m must satisfy the following inequality:

$$m \ge \text{ck} \log(N/k) \tag{3}$$

where c is a constant that depends on each particular instance and k is the order of sparsity. It has been shown in [23] that the low coherence between $\mathbf{\Phi}$ and $\mathbf{\Psi}$ is guaranteed by choosing a sensing matrix with random entries and the incoherence can be directly achieved using a Gaussian sensing matrix. As a result, there is no need to check incoherence.

B. Compressive Sensed Band Domain

Due to acquisition by hundreds of contiguous spectral bands, a hyperspectral imagery generally has an enormous data volume, which contains wealthy spectral information. Accordingly, CS is quite suited for hyperspectral imaging. Over the past years, sparsity has been used as an additional constraint to expand the existing hyperspectral imaging algorithms in various applications, such as linear spectral unmixing [26]–[28], BSel [29], feature extraction [30], [31], and classification [32], [33]. Interestingly, the constraint of sparsity is basically used for signal recovery. However, unlike communication/signal processing that generally requires signal reconstruction or recovery, many applications in hyperspectral data exploitation do not need this requirement since only end processed results are of interest. In particular, many subtle material substances can be uncovered by a hyperspectral imaging sensor and many of them can be background signatures or interferers. It does not make sense to recover these unwanted signal sources. In addition, since hyperspectral data generally have redundant spectral information resulting from close interband correlation, compressing bands is crucial and critical in hyperspectral data exploitation. Thus, several recent works have investigated this issue in how to process hyperspectral images in CSBD [34]–[36].

More specifically, we can consider a hyperspectral image, $\text{HSI} \in \Re^{N_x \times N_y \times L}$, as a data cube with two spatial dimensions N_x and N_y and a spectral dimension L. A CSBD image, $\text{HSI}_{\text{CSBD}} \in \Re^{N_x \times N_y \times m}$, can be formed by sensing m linear mixtures of the original spectral bands. The compressed acquisition of single hyperspectral pixel vector $\mathbf{r} \in \Re^{L \times 1}$ can be expressed as

$$\bar{\mathbf{r}} = \mathbf{\Phi}\Psi\mathbf{r} + \mathbf{n} \tag{4}$$

where $\bar{\mathbf{r}} \in \Re^{m \times 1}$ is the compressed measurement vector, $\mathbf{\Phi} \in \Re^{m \times L}$ is the compressed sampling matrix, $\mathbf{\Psi} \in \Re^{L \times L}$ is the sparse representation basis, and $\mathbf{n} \in \Re^{m \times 1}$ is a noise term. The model in (4) requires that the signals are sparse and sampled incoherently [19]. The challenge in designing a CS system is choosing $\mathbf{\Phi}$ and $\mathbf{\Psi}$ to satisfy these requirements

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

for a given application. A previous work [34] showed that a hyperspectral model, well suited for analysis in the band compressed domain, can be developed by using random sampling matrices. More specifically, $\mathbf{\Phi}$ is generated from any random distribution that satisfies the concentration of measure inequality [21]. Such random sampling matrices have been shown to be incoherent with any representation basis, $\mathbf{\Psi}$, and also satisfy the restricted isometry property (RIP) that is often desired in CS applications. RIP guarantees that relative measures of distance are preserved in the compressively sensed space, effectively making the sampling matrix an orthonormal projector $\mathbf{\Phi}\mathbf{\Phi}^T \approx \mathbf{I}$. Finally, when such a random sensing strategy is used, the sparse basis, $\mathbf{\Psi}$, only needs to be identified for sparse recovery. This "universality" property of random sensing also guarantees that all information is captured in CSBD for a sufficient number of CSBs [21]. Consequently, (4) can be further reduced to a simplified model as shown in the following:

$$\bar{\mathbf{r}} = \mathbf{\Phi}_m \mathbf{r} + \mathbf{n} \tag{5}$$

where $\mathbf{\Phi}_m$ represents a random sampling matrix using m CSBs.

One innovative concept of CSBD is to operate the sensing matrix $\mathbf{\Phi}$ on bands not samples. It is (5) that applies $\mathbf{\Phi}$ to reduce the entire band dimensionality L to m via $\mathbf{\Phi}_m$. It should be noted that the samples in Section II-A and bands in Section II-B resulting from the sensing matrix $\mathbf{\Phi}$ are indeed linear mixtures by $\mathbf{\Phi}$. There are two key differences between CS and CSBD. One is that CSBD operates the sensing matrix $\mathbf{\Phi}$ on spectral bands as opposed to CS that operates $\mathbf{\Phi}$ on signal samples. A second difference is that CS requires both $\mathbf{\Phi}$ for sensing and $\mathbf{\Psi}$ for signal recovery, whereas CSBD only requires $\mathbf{\Phi}$ for sensing bands without specifying $\mathbf{\Psi}$ because of the universality property in (5).

III. Band Sampling

The CS theory discussed in Section II-A requires sparsity and incoherence to make CS work. The CSBD discussed in Section II-B removes the requirement of sparsity without signal recovery via the universality model in (5) in which case only the number of sensed bands m matters. This m is equivalent to n_{BSam} in (1) used by BSam. This also indicates that both signal sampling and BSam share the same key issue, which is the determination of sampling rate.

For a given n_{BSam}, the maximal incoherence can be achieved by UBSam, provided that the band coherence is interpreted by interband correlation. This is exactly the same way that the Nyquist rate is implemented uniformly in signal sampling and so is UBSel used in BSel. Next, we look into the issue of sparsity that is closely related to band representation in our case. In CSBD, this property is implicitly taken care of by the universality model in (5), in which case there is no need of specifying the sparse matrix, $\mathbf{\Psi}$ used for signal recovery. For BSam, how this can be taken care of implicitly?

First of all, we consider the entire bands as the entire data space, where each band is considered as a signal with full representation. Then, the number of sampled bands n_{BSam} can be considered as band sparsity corresponding to the sparsity order k in (3) used to express a signal by its sparse representation with only a few nonzero components, that is, for a given application, a set of appropriately sampled n_{BSam} bands should faithfully represent the entire bands in terms of (5). Second, we may take advantage of the idea of using SFs by ITS developed in [25] to capture spatial information from the processed data by replacing training samples with n_{BSam} sampled bands to yield a version of IBSam that can accomplish what the data sparse representation does in CS. The details of implementing IBSam at the jth iteration can be described in Algorithm 1.

Algorithm 1 IBSam

1. Initial condition: let $\{\text{SF}_m\}_{m=1}^M$ be a set of M spatial filters and randomly sample n_{BSam} bands from original full spectral band set, $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^L$, denoted by $\mathbf{\Omega}_{\text{BSam}}^0 = \left\{\mathbf{B}_{l_k}^0\right\}_{k=1}^{n_{\text{BSam}}}$. Set $\mathbf{\Omega}_{\text{SFMaps}}^0 = \emptyset$ to be an empty set.
2. At the jth iteration with $j \geq 1$,
   (a) process $\mathbf{\Omega}_{\text{total}}^j$ to produce the processed data $\text{PD}^{(j)}$ by a processor or a filter or more generally an operator (to be specified by applications such as detection maps by a detector, classification maps by a classifier).
   (b) Implement $\{\text{SF}_m\}_{m=1}^M$ on $\text{PD}^{(j)}$ obtained in step 2 to capture the spatial information among the data samples in $\text{PD}^{(j)}$ and feeds this obtained spatial information, denoted by $\left\{\text{SFMap}_m^j\right\}_{m=1}^M$, back to $\text{PD}^{(j)}$ by a feedback loop.
   (c) Form
   $$\begin{aligned}\mathbf{\Omega}_{\text{SFMaps}}^j &= \mathbf{\Omega}_{\text{SFMaps}}^{j-1} \cup \left\{\text{SFMap}_m^j\right\}_{m=1}^M \\ &= \bigcup_{i=1}^{j} \left\{\text{SFMap}_m^i\right\}_{m=1}^M \end{aligned} \tag{6}$$
   (d) Randomly sample bands from the original full spectral band set, $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^L$ according to n_{BSam}, denoted by $\mathbf{\Omega}_{\text{BSam}}^j = \left\{\mathbf{B}_{l_k}^j\right\}_{k=1}^{n_{\text{BSam}}}$ to form a new data set, $\mathbf{\Omega}_{\text{total}}^{j+1} = \mathbf{\Omega}_{\text{SFMaps}}^{j-1} \cup \mathbf{\Omega}_{\text{BSam}}^j$.
   (e) Check if a stopping rule is satisfied. The criterion to be used is the Tanimoto index (TI) defined in [37] by
   $$\text{TI}^{(j)} = \frac{|A^{(j-1)} \cap A^{(j)}|}{|A^{(j-1)} \cup A^{(j)}|} \tag{7}$$
   where $|A|$ is the size of set A, $A^{(j)}$ and $A^{(j-1)}$ are results obtained the jth and $(j-1)^{\text{st}}$ iterations. When TI of the jth iteration is greater than a prescribed threshold ε, the iteration is terminated. Otherwise, continue.
   (f) Let $j \leftarrow j+1$ and go to step 2(a).

It should be noted that $\mathbf{\Omega}_{\text{SFMaps}}^{j-1}$ is only made up of spatial filtered maps and does not include BSam-sampled bands. Also, let $\mathbf{\Omega}_{\text{BSam}}^j = \{\mathbf{B}_{l_k}\}_{k=1}^{n_{\text{BSam}}}$ be the set of bands sampled by BSam at the jth iteration and $\mathbf{\Omega}_{\text{total}}^j$ be the data cube at the jth iteration.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

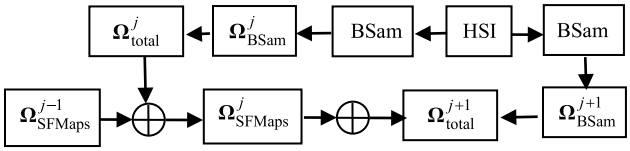


Fig. 1. Block diagram of IBSam at the jth iteration.

Fig. 1 shows a simple block diagram to illustrate the above iterative process.

It is also worth noting that two types of ITS developed in [25], iterative fixed training sampling (IFTS) and iterative random training sampling (IRTS), can also be applied to IBSam to yield IFBSam and IRBSam.

In order to equip BSam with ability of obtaining spatial information, we need to incorporate a set of SFs in BSam to derive IBSam, which can include the SF-captured information from the processed data iteratively.

Next, to take care of incoherence, two different ideas are explored for IBSam to maximize incoherence.

1) One is referred to as IUBSam based on information theory, which assumes that uniform distribution yields the maximal entropy.
2) A second one is IRBSam in correspondence to random sampling based on random sensing matrix $\mathbf{\Phi}$ from the CS theory.

Both IUBSam and IRBSam will be discussed in great detail in Sections IV and V, respectively.

IV. Uniform Band Sampling

The simplest way to sample bands in order to achieve maximal incoherence is UBSam similar to the uniform sampling by the Nyquist rate. With the assumption that each sampled band is considered equally significant, UBSam should give the maximum possible incoherence resulting from the maximum entropy achieved by the uniform distribution. This greatly reduces the problem because it implies that only the number of bands n_{BSam} needs to be determined rather than a specific sampled band subset. In addition, the problem complexity can be significantly reduced to remove the computational burdens that would otherwise limit the practicality of an exhaustive technique. In this case, UBSam can be modified from IBSam in Fig. 1 described in Algorithm 2.

Since UBSam is determined by its initial sampling of the first band, there are n_{UBSam} different selections $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ for the initial sampling selection. Two scenarios can be implemented for UBSam.

A. Exhaustive UBSam

To best utilize all possible n_{UBSam} different initial sampling selections, we can run UBSam iteratively through all possible n_{UR} initial sampling selections, $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ in step 1 to yield an exhaustive version of UBSam, called EUBSam. In this case, step 2(e) in UBSam will be terminated after n_{UBSam}. Once EUBSam is terminated, EUBSam exhausts all the n_{UBSam} initial sampling selections and the final results are the results always obtained by using all information provided by previous iterations.

Algorithm 2 UBSam

1. Initial condition:
   Let n_{UBSam} be the number of bands to be sampled uniformly.
   Let the original full spectral band set, $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$.
   Uniformly sample n_{UBSam} bands from $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$, denoted by $\mathbf{\Omega}_{\text{UBSAM}} = \{\mathbf{B}_{l_k}\}_{k=1}^{n_{\text{UBSAM}}}$.
2. At the jth iteration with $j \geq 1$, the following iterative process.
   (a) Set $\mathbf{\Omega}^{j}_{\text{UBSAM}} = \mathbf{\Omega}_{\text{UBSAM}}$ and let $\mathbf{\Omega}^{j}_{\text{total}} = \mathbf{\Omega}^{j}_{\text{UBSAM}} \cup \mathbf{\Omega}^{j-1}_{\text{SFMaps}}$.
   (b) Process $\mathbf{\Omega}^{j}_{\text{total}}$ to produce $\text{PD}^{(j)}$.
   (c) Implement a set of M spatial filters $\{\text{SF}_m\}_{m=1}^{M}$ on $\text{PD}^{(j)}$ obtained in (a) to capture the spatial correlation among processed data samples and feed this obtained spatial information back by a feedback loop to the currently being processed data set, denoted by $\{\text{SFMap}_m^j\}_{m=1}^{M}$.
   (d) Form

$$\mathbf{\Omega}^{j+1}_{\text{SFMaps}} = \mathbf{\Omega}^{j}_{\text{SFMaps}} \cup \{\text{SFMap}_m^j\}_{m=1}^{M} = \bigcup_{i=1}^{j} \{\text{SFMap}_m^i\}_{m=1}^{M} \tag{8}$$

   (e) Check $j = 1$. If yes, go to step 2. Otherwise, check if a stopping rule defined by (7) is satisfied. If yes, the algorithm is terminated. Otherwise, continue.
   (f) Let $j \leftarrow j + 1$ and go to step 2(a). That is, repeat the entire iterative cycle (a)–(f).

B. Random UBSam

Unlike EUBSam, which must run through all possible n_{UBSam} initial sampling selectins in $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$, another scenario is to randomly select a uniform sampling selection from $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ in step 1 of UBSam.

There are three differences between EUBSam and RUBSam.

1) EUBSam exhausts all the uniform sampling selections in $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$, whereas RUBSam does not and it may be terminated before n_{UBSam} iterations.
2) EUBSam must run all the uniform sampling selections in $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$. By contrast, RUBSam may run one uniform sampling selection more than once and some other uniform sampling selection may never be selected.
3) EUBSam will automatically be terminated after n_{UBSam} iterations. However, RUBSam requires a stopping rule to terminate its algorithm.

C. Fixed UBSam

As a special case of RUBSam and EUBSam, an FUBSam is developed to randomly select one from $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ as RUBSam does, but it differs from RUBSam in the sense that FUBSam stays with its selected initial sampled bands during the entire iterative process as opposed to RUBSam that randomly resamples bands from $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ at every

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

Algorithm 3 RUBSam

1. Initial condition:
   Let n_{UBSam} be the number of bands to be sampled uniformly.
   Let the original full spectral band set, $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$.
   Let $\left\{\text{UBSam}_j^{\text{initial}}\right\}_{j=1}^{n_{\text{UBSam}}}$ be all possible n_{UBSam} initial sampling selectins.
   Randomly select an initial sampling selection from $\left\{\text{UBSam}_j^{\text{initial}}\right\}_{j=1}^{n_{\text{UBSam}}}$, denoted by $\mathbf{\Omega}_{\text{UBSAM}}^{0} = \left\{\mathbf{B}_{l_k}^{0}\right\}_{k=1}^{n_{\text{UBSAM}}}$
2. At the jth iteration with $j \geq 1$,
   (a) Randomly select an initial sampling selection from $\left\{\text{UBSam}_j^{\text{initial}}\right\}_{j=1}^{n_{\text{UBSam}}}$, denoted by $\mathbf{\Omega}_{\text{UBSAM}}^{j} = \left\{\mathbf{B}_{l_k}^{j}\right\}_{k=1}^{n_{\text{UBSAM}}}$ and let $\mathbf{\Omega}_{\text{total}}^{j} = \mathbf{\Omega}_{\text{UBSAM}}^{j} \cup \mathbf{\Omega}_{\text{SFMaps}}^{j-1}$.
   (b) Process $\mathbf{\Omega}_{\text{total}}^{j}$ to produce $\text{PD}^{(j)}$.
   (c) Implement a set of M spatial filters $\{\text{SF}_m\}_{m=1}^{M}$ on $\text{PD}^{(j)}$ obtained in (a) to capture the spatial correlation among processed data samples and feed this obtained spatial information back by a feedback loop to the currently being processed data set, denoted by $\left\{\text{SFMap}_m^{j}\right\}_{m=1}^{M}$.
   (d) Form

$$\begin{aligned}\mathbf{\Omega}_{\text{SFMaps}}^{j+1} &= \mathbf{\Omega}_{\text{SFMaps}}^{j} \cup \left\{\text{SFMap}_m^{j}\right\}_{m=1}^{M} \\ &= \bigcup_{i=1}^{j} \left\{\text{SFMap}_m^{i}\right\}_{m=1}^{M} \end{aligned} \tag{9}$$

   (e) Check $j = 1$. If yes, go to step 2. Otherwise, continue.
   (f) Check a stopping rule defined by (7). If it is not satisfied, let $j \leftarrow j + 1$ and randomly select one uniform sampling selection from $\left\{\text{UBSam}_j^{\text{initial}}\right\}_{j=1}^{n_{\text{UBSam}}}$, denoted by $\text{UBSam}_j^{\text{initial}}$. Let $\mathbf{\Omega}_{\text{UBSAM}}^{j} = \text{UBSam}_j^{\text{initial}}$ and go to step 2(a).

iteration. Nevertheless, they both use the same stopping rule to terminate their algorithms.

On the other hand, FUBSam is also different from EUBSam in the sense that it only runs one selection from $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ through its iterative process, while EUBSam runs through all selections in $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ without a stopping rule because it exhausts $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$.

Finally, it is interesting to note that as for BSel, UBSel is generally determined by how the first band is selected. Thus, it can be considered to be a special case of FUBSam as an initial condition without an iterative process. On the other hand, since there are n_{UBSam} different initial sampling selections that can be used for UBSam, the results vary with different initial sampling selections. EUBSam avoids this issue by taking advantage of its iterative process to run through all possible n_{UBSam} initial sampling selections.

V. Iterative Random Band Sampling

The UBSam described in Section IV assumes that UBSam maximizes incoherence among selected bands by the uniform distribution. However, if the signal sampling rate is below the Nyquist rate, uniform signal sampling will cause signal aliasing. To deal with this dilemma, the CS theory introduces random signal sampling to resolve this issue. Using the same token, if it is not known whether VD-estimated n_{BSam} is above the true BSamR in (1) or not, the best strategy is to use RBSam that is analogous to random signal sampling considered in CS. Unlike FUBSam which samples bands as its initial condition and then fixes the same initial set of sampled bands during the entire iterative process, IRBSam presented in this section is to sample bands randomly at each iteration.

To be more specific, we can follow a similar idea to develop RBSam by randomly sampling n_{BSam} bands at every iteration instead of running each initial sampling selection in $\{\text{UBSam}_j^{\text{initial}}\}_{j=1}^{n_{\text{UBSam}}}$ as UBSam does. The difference between IRBSam and UBSam is that the former requires a stopping rule to terminate its iterative process, whereas the latter only needs to run n_{UBSam} times.

Its detailed implementation is given in Algorithm 4.

Algorithm 4 IRBSam

1. Initial condition:
   Let n_{BSam} be given.
   Set $\mathbf{\Omega}_{\text{SFMaps}}^{0} = \emptyset$.
   Let the original full spectral band set, $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$.
   Uniformly sample n_{BSam} from $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$, denoted by $\mathbf{\Omega}_{\text{RBSam}}^{0} = \left\{\mathbf{B}_{l_k}^{0}\right\}_{k=1}^{n_{\text{BSam}}}$
2. At the jth iteration with $j \geq 1$, the following iterative process.
   (a) Randomly sample n_{BSam} bands from $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$, denoted by $\mathbf{\Omega}_{\text{RBSam}}^{j} = \left\{\mathbf{B}_{l_k}^{j}\right\}_{k=1}^{n_{\text{BSam}}}$. Let $\mathbf{\Omega}_{\text{total}}^{j} = \mathbf{\Omega}_{\text{RBSam}}^{j} \cup \mathbf{\Omega}_{\text{SFMaps}}^{j-1}$.
   (b) Process $\mathbf{\Omega}_{\text{total}}^{j}$ to produce $\text{PD}^{(j)}$.
   (c) Implement a set of M spatial filters $\{\text{SF}_m\}_{m=1}^{M}$ on $\text{PD}^{(j)}$ obtained in (a) to capture the spatial correlation among spectral processed data samples and feed this obtained spatial information back by a feedback loop to the currently being processed data set, denoted by $\left\{\text{SFMap}_m^{j}\right\}_{m=1}^{M}$.
   (d) Form

$$\begin{aligned}\mathbf{\Omega}_{\text{SFMaps}}^{j+1} &= \mathbf{\Omega}_{\text{SFMaps}}^{j} \cup \left\{\text{SFMap}_m^{j}\right\}_{m=1}^{M} \\ &= \bigcup_{i=1}^{j} \left\{\text{SFMap}_m^{i}\right\}_{m=1}^{M} \end{aligned} \tag{10}$$

   (e) Check $j = 1$. If yes, go to step 2. Otherwise, check if a stopping rule is satisfied. If yes, the algorithm is terminated. Otherwise, continue.
   (f) Let $j + 1 \leftarrow j$ and go to step 2(a). That is, repeat the entire iterative cycle (a)-(g).

VI. Iterative Fixed Band Sampling

In step 2(a) of IRBSam developed in Section V, the bands at the jth iteration are sampled randomly. This idea is derived

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

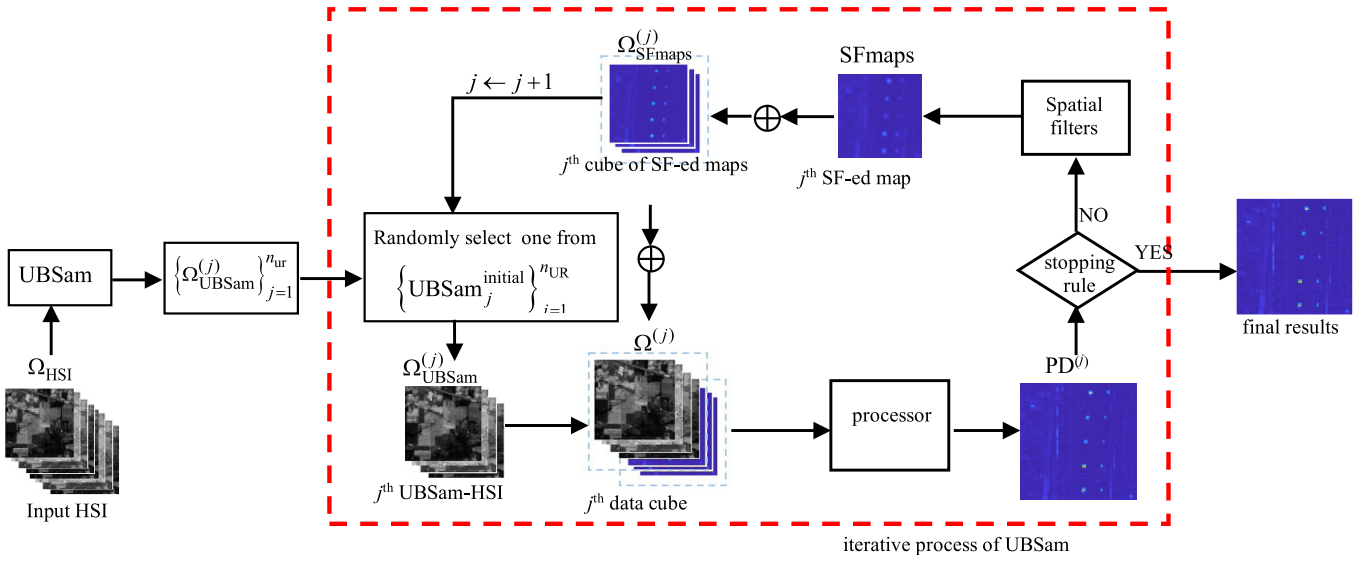


Fig. 2. Flowchart of RUBSam.

Algorithm 5 IFBSam

1. Initial condition:
   Let n_{BSam} be given.
   Set $\mathbf{\Omega}^0_{\mathrm{SFMaps}} = \emptyset$.
   Let the original full spectral band set, $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$.
   Randomly sample n_{BSam} bands from $\mathbf{\Omega} = \{\mathbf{B}_l\}_{l=1}^{L}$, denoted by $\mathbf{\Omega}^0_{\mathrm{RBSam}} = \{\mathbf{B}^0_{l_k}\}_{k=1}^{n_{\mathrm{BSam}}}$.
2. At the jth iteration with $j \geq 1$, the following iterative process.
   (a) Set $\mathbf{\Omega}^j_{\mathrm{FBSam}} = \mathbf{\Omega}^0_{\mathrm{RBSam}}$ and let $\mathbf{\Omega}^j_{\mathrm{total}} = \mathbf{\Omega}^j_{\mathrm{FBSam}} \cup \mathbf{\Omega}^{j-1}_{\mathrm{SFMaps}}$.
   (b) Process $\mathbf{\Omega}^j_{\mathrm{total}}$ to produce $\mathrm{PD}^{(j)}$.
   (c) Implement a set of M spatial filters $\{\mathrm{SF}_m\}_{m=1}^{M}$ on $\mathrm{PD}^{(j)}$ obtained in (a) to capture the spatial correlation among processed data samples and feed this obtained spatial information back by a feedback loop to the currently being processed data set, denoted by $\{\mathrm{SFMap}^j_m\}_{m=1}^{M}$.
   (d) Form

$$\mathbf{\Omega}^{j+1}_{\mathrm{SFMaps}} = \mathbf{\Omega}^j_{\mathrm{SFMaps}} \cup \{\mathrm{SFMap}^j_m\}_{m=1}^{M} = \bigcup_{i=1}^{j} \{\mathrm{SFMap}^i_m\}_{m=1}^{M} \tag{11}$$

   (d) Check $j = 1$. If yes, go to step 2. Otherwise, check if a stopping rule specified by (7) is satisfied. If yes, the algorithm is terminated. Otherwise, continue.
   (f) Let $j \leftarrow j+1$ and go to step 2(a). That is, repeat the entire iterative cycle (a)–(f)

from RTS. As a result, the sampled bands processed in every iteration are different. Interestingly, as opposed to RTS, there is FTS developed in [25], which can also be used to yield IFBSam. Its idea follows IRBSam except $\mathbf{\Omega}^j_{\mathrm{RBSam}} = \{\mathbf{B}^j_{l_k}\}_{k=1}^{n_{\mathrm{BSam}}}$ replaced by $\mathbf{\Omega}^j_{\mathrm{FBSam}} = \{\mathbf{B}^0_{l_k}\}_{k=1}^{n_{\mathrm{BSam}}}$ for all $j \geq 1$ where the same initial set of sampled bands is fixed and used for all subsequent iterations, i.e., $\mathbf{\Omega}^j_{\mathrm{FBSam}} = \mathbf{\Omega}^0_{\mathrm{FBSam}} = \{\mathbf{B}^0_{l_k}\}_{k=1}^{n_{\mathrm{BSam}}}$ for all $j \geq 1$. The step-by-step implementation of IFBSam is summarized in Algorithm 5.

There are two special cases that can be derived from IFBSam.

1) One is FUBSam where the initial condition $\{\mathbf{B}^0_{l_k}\}_{k=1}^{n_{\mathrm{BSam}}}$ used in step 1 of IFBSam is replaced by one uniform sampling selection from $\{\mathrm{UBSam}^{\mathrm{initial}}_j\}_{j=1}^{n_{\mathrm{UR}}}$.
2) Another is BSel where BSel can be implemented as an initial condition of IFBSam without using SFs to perform an iterative process. In this case, BSel can be viewed as FBSam with no iterative process implemented.

Figs. 2–5 also show the graphic block diagrams of detailed implementations of RUBSam, EUBSam, IRBSam, and IFBSam, respectively.

Finally, we can summarize the relationships among all various versions of BSam: 1) UBSam, which includes RUBSam, FUBSam, and EUBSam, and 2) IBSam, which includes IRBSam and IFBSam in the following diagram:

$$\begin{cases} \mathrm{BSam} \Rightarrow \\ \quad \mathrm{UBSam} \Rightarrow \begin{cases} \mathrm{RUBSam\ (Fig.\ 2)} \\ \mathrm{EUBSam\ (Fig.\ 3)} \\ \mathrm{FUBSam} \Rightarrow \mathrm{UBSel} \end{cases} \\ \quad \mathrm{IBSam} \\ \quad (\mathrm{Fig.1}) \Rightarrow \begin{cases} \mathrm{IRBSam\ (Fig.\ 4)} \\ \mathrm{IFBSam} \Rightarrow \begin{cases} \mathrm{IFUBSam} \\ \mathrm{(Fig.\ 5)} \\ \mathrm{BSel.} \end{cases} \end{cases} \end{cases} \tag{12}$$

VII. BSam Rate Estimated by VD

Upon implementing BSam, two parameters are needed to be addressed.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

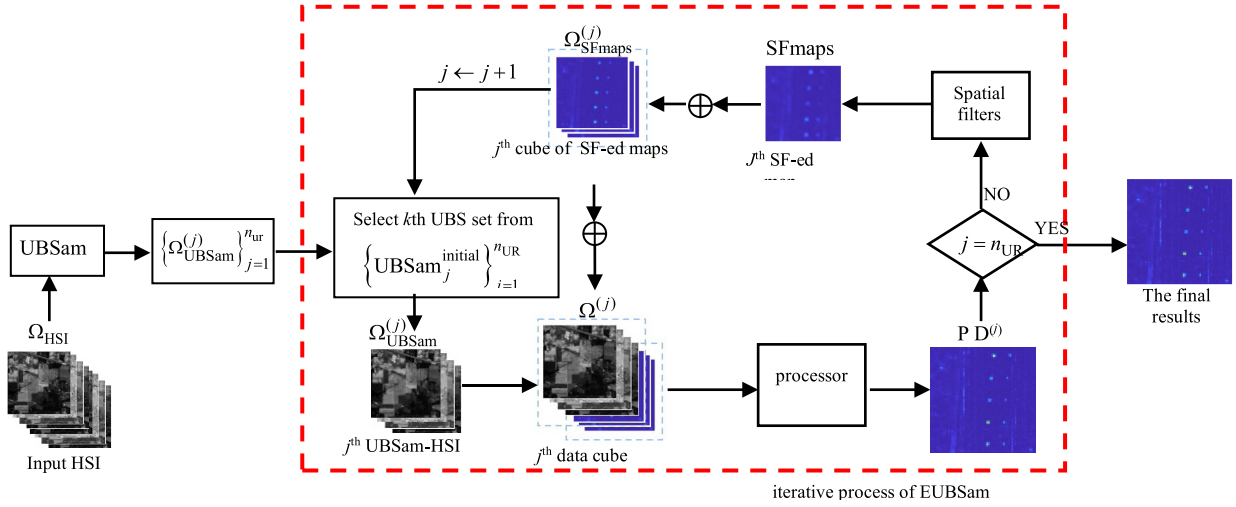


Fig. 3. Flowchart of EUBSam.

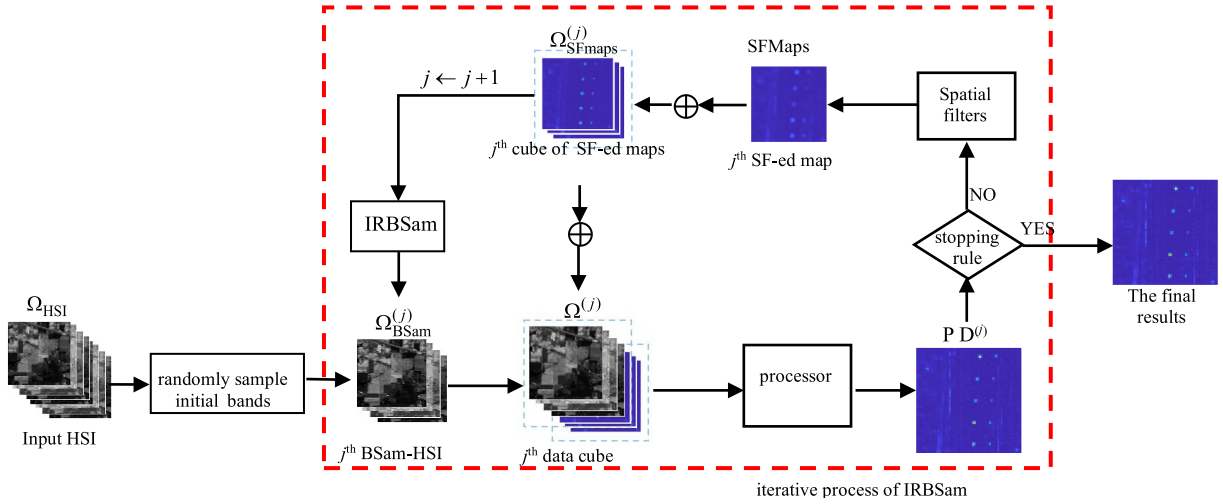


Fig. 4. Flowchart of IRBSam.

1) One is the parameter M that is the number of SFs to be used by BSam generally determined by different applications. For example, if the application is a single target detection or anomaly detection, M is set to 1. When the application is a p-class classification, M is set to the number of classes of interest, which is p.
2) The second but key parameter is BSamR defined in (1), i.e., n_{BSam}, which can be estimated by VD to be discussed in Section VII. This is similar to the signal sampling rate determined by the Nyquist rate in communications/signal processing.

A natural assumption made on hyperspectral imagery is that the number of spectrally distinct signatures present in hyperspectral imagery, denoted by n_s, is generally much smaller than the total number of spectral bands, L spectral bands, i.e., $n_s \ll L$. Thus, the idea of VD developed in [19] and [20] is derived from two basic principles, pigeonhole principle [38] and orthogonality principle [39] described in [40, Ch. 1]. More specifically, the pigeonhole principle guarantees that for each spectral signature, there exists at least one band that can be used to specify and accommodate this particular spectral signature. This is equivalent to saying that BSam can sample an appropriate band to characterize a given spectral signature. On the other hand, the orthogonality principle guarantees that once a band is sampled, this band cannot be used again to specify any other spectral signature. In other words, no single band can be used to specify two spectrally distinct signatures. As a result of these two principles, VD can be used to determine n_s.and, thus, in turn to determine n_{BSam}.

Based on the pigeonhole principle and orthogonality principle, an effective method for estimating VD was developed by Harsanyi *et al.* [41], commonly referred to as Harsanyi–Farrand–Chang (HFC) method along with its MATLAB codes also available in [40, pp. 997–998] where one pigeonhole is specified by one particular spectral band and each pigeon is specified by one signature. It then formulates the VD estimation problem as a binary composite hypothesis testing problem where the null hypothesis H_0 and alternative hypothesis H_1 are specified by covariance matrix obtained eigenvalues $\{\lambda_l\}_{l=1}^{L}$ and correlation matrix obtained eigenvalues $\{\tilde{\lambda}_l\}_{l=1}^{L}$ to represent the absence and presence of a signature, respectively. More precisely, for each spectral band l, a binary composite hypothesis testing problem can be cast as follows:

$$H_0(\text{signature absence}): p_0(z_l) \approx \tilde{\lambda}_l - \lambda_l = 0$$
$$\text{versus}$$
$$H_1(\text{signature presence}): p_1(z_l) \approx \tilde{\lambda}_l - \lambda_l > 0. \quad (13)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

A Neyman–Pearson detector of $z_l = \hat{\lambda}_l - \lambda_l$, denoted by $\delta^{\mathrm{NP}}(z_l)$ for (12), can be obtained by maximizing the detection power $\mathrm{P_D}$ subject to the false alarm probability $\mathrm{P_F}$ being fixed at a given value, β [39]. The level of significance for the detector is specified by β, which determines the threshold value τ_l in the following randomized decision rule:

$$\delta^{\mathrm{NP}}(z_l) = \begin{cases} 1; & \text{if } \Lambda(z_l) > \tau_l \\ 1 \text{ with probability } \kappa; & \text{if } \Lambda(z_l) = \tau_l \\ 0; & \text{if } \Lambda(z_l) < \tau_l \end{cases} \tag{14}$$

where the likelihood ratio test $\Lambda(z_l)$ is given by $\Lambda(z_l) = p_1(z_l)/p_0(z_l)$ with $p_0(z_l)$ and $p_1(z_l)$ being probability distributions under H_0 and H_1 for (14), respectively. Thus, a case of $z_l = \hat{\lambda}_l - \lambda_l > \tau_l$ indicates that $\delta^{\mathrm{NP}}(z_l)$ in (14) fails the test, which means that there is signal energy assumed to contribute to the eigenvalue, $\hat{\lambda}_l$, in the lth spectral band. It should be noted that the test (14) must be performed for each of L spectral bands. Therefore, for each pair of $\hat{\lambda}_l - \lambda_l$, the threshold τ_l varies with the fixed value of β and is spectral band-dependent. Using (14), VD can be determined by calculating

$$\mathrm{VD}^{\mathrm{NP}}_{\mathrm{HFC}}(\mathrm{P_F} \le \beta) = \sum_{l=1}^{L} \left\lfloor \delta^{\mathrm{NP}}(z_l) \right\rfloor \tag{15}$$

with a predetermined false alarm probability $\mathrm{P_F} = \beta$, where $\lfloor \delta(z_l) \rfloor = 1$ only if $\delta^{\mathrm{NP}}(z_l) = 1$ and $\lfloor \delta(z_l) \rfloor = 0$ if $\delta^{\mathrm{NP}}(z_l) < 1$. By virtue of (15), we can estimate n_s by

$$n_{\mathrm{BSam}}(\beta) = \mathrm{VD}^{\mathrm{NP}}_{\mathrm{HFC}}(\mathrm{P_F} \le \beta). \tag{16}$$

It is the constraint value β in (15) that determines n_s and varies with different applications.

Furthermore, the HFC method was later extended in [19] and [20] referred to as the noise-whitened HFC (NWHFC) method, which whitens the data prior to the HFC method. The resulting VD is defined as $\mathrm{VD}^{\mathrm{NP}}_{\mathrm{NWHFC}}(\mathrm{P_F} \le \beta)$. More details about VD can be found in [19] and [20].

Finally, we should point out that the effectiveness of BSam is determined by its n_{BSam} with the constraint parameter β. This is similar to the sparsity order, k in (3) used by CS. If n_{BSam} is underestimated by VD, RBSam can be used for compensation.

VIII. Generalizability of BSam

In analogy with CS that does not specify particular signal processors, BSam also does not specify hyperspectral imaging processors or a particular set of M SFs used by BSam. These two elements can be determined by different applications. As a consequence, BSam has great potential in its generalizability. In what follows, we describe several key elements in BSam that can be generalized in various directions.

1) BSam reinvents the wheel by taking advantage of a recently developed IFTS/IRTS in [25] to improve BSel. It utilizes a set of M SFs to capture spatial information from processed data on sampled bands and then iteratively feeds the spatial filtered information back to expand the currently being processed data cube to create a new data cube for the next round iteration by BSam. More precisely, the key idea is to perform random sampling on bands rather than training samples as IFTS/IRTS does in [25]. Interestingly, performing IRBSam on the expanded data cube by adding spatial filtered information to the current data cube via iterative loops can do better than many existing BSel methods.
2) Since we can always start with the bands selected by any BSel method as an initial condition of BSam, BSel is indeed a special case of BSam, where BSel is considered as BSam, which only implements its initial condition without taking advantage of SFs to further carry out an iterative process. In particular, BSel can be considered as FBSam without iterative process.
3) Unlike conventional BSel methods which select specific and deterministic bands, the bands sampled by IRBSam are completely random and not reproducible. However, this is not an issue since the end results are what we care about and are determined by applications. It is immaterial with which bands are sampled. This is exactly the same as random sampling that CS uses to generate random sensing matrices without referring to sampled signals that are random and cannot be reproducible either.
4) The utility of BSam is determined by its processed data, which can be specified by a particular application. To illustrate its versatility in various applications, three specific tasks, target detection, anomaly detection, and classification, which have received considerable interest in hyperspectral data exploitation, are studied.
5) There are a variety of SFs that are available and can be used in conjunction with BSam, for example, bilateral filters guided filters, Gaussian filters, and Gabor filters.

IX. Applications to Target/Anomaly Detection

Generally speaking, different applications require use of different imaging processors. BSam allows users to specify a custom-designed processor to meet various applications. In this section, we present three applications, hyperspectral target detection with the processor specified by CEM developed in [42] and [43] and its improved version, hierarchical CEM (H-CEM) [44]. Although there is another improved CEM, called ensemble-based cascaded constrained energy minimization (E-CEM) also developed in [45], its performance is very close to H-CEM, so are BSam. In this case, only CEM and H-CEM were implemented for experiments.

It has been shown in [9] that constrained target band selection (CTBS)-based methods, BF-MinV-BP, BF-MaxV-BP, BF-SF-CTBS, and BF-SB-CTBS, performed better than most existing BSel methods for target detection. They were implemented for study. In addition, a popular constrained band selection (CBSel) [3] along with UBSel was also implemented for comparative study and analysis because they both are also independent of applications such as BSam. Nevertheless, it should be noted that other BS methods can also be used for this purpose as well. Due to the limited space, other BSel methods were included in this article but can be referred to in [3].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

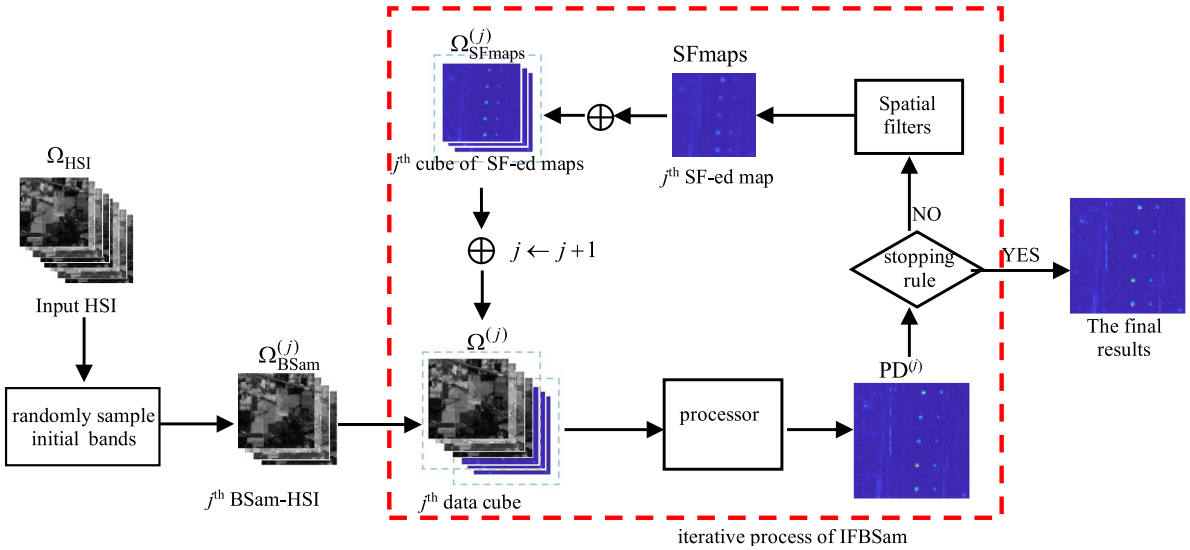


Fig. 5. Flowchart of IFBSam.

A. 3-D ROC Curve-Derived Detection Measures

In hyperspectral target detection, a commonly used tool is 2-D receiver operating characteristic (ROC) curve, which plots a function of detection probability P_D versus false alarm probability P_F of a detector. The area under curve (AUC), denoted by $\text{AUC}_{(D,F)}$, is generally used to measure its detection performance. Unfortunately, recent works in [46] have shown that solely relying on $\text{AUC}_{(D,F)}$ was misleading in many cases for hyperspectral target detection. This is mainly caused by the fact that both P_D and P_F are calculated by the same threshold τ used by the detector. As a result, when both P_D and P_F are very high, its $\text{AUC}_{(D,F)}$ is also very high. Similarly, when both P_D and P_F are very low, its $\text{AUC}_{(D,F)}$ is also very low. In other words, P_D and P_F cannot work alone, and thus, P_D and P_F cannot individually measure target detectability (TD) and BS independently. To address this issue, an effective 3-D ROC analysis-based evaluation tool was developed in [46], which extends the traditional 2-D ROC analysis by including the threshold τ as an additional parameter to represent a 3-D ROC curve as a function of three parameters, P_D, P_F, and τ. As a result, a 3-D ROC curve is a plot of a triplet parameter vector (P_D,P_F,τ) from which three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) with their respective AUC values, denoted by, $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$ can be generated where the traditional 2-D ROC curve becomes a part of 3-D ROC curve. For those who are interested in using the 3-D ROC analysis for detection performance evaluation, a MATLAB-based software package of how to generate a 3-D ROC curve is available on the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), University of Maryland, Baltimore County (UMBC) [47].

Thus, in addition to $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$ generated by a 3-D ROC curve, five more detection measures developed in [46] can be further derived as follows.

1) TD is denoted by AUC_{TD} and defined by

$$0 \le \text{AUC}_{\text{TD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)}) \le 2. \quad (17)$$

2) BKG suppression measure is denoted by AUC_{BS} and defined by

$$-1 \le \text{AUC}_{\text{BS}} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)}) \le 1. \quad (18)$$

3) Target detection and BS (TDBS) are denoted by AUC_{TDBS} and defined by

$$-1 \le \text{AUC}_{\text{TDBS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)}) \le 1. \quad (19)$$

4) Overall detection probability (ODP) is denoted by AUC_{ODP} and defined by

$$-1 \le \text{AUC}_{\text{ODP}} = \text{AUC}_{(D,F)}+\text{AUC}_{(D,\tau)}-\text{AUC}_{(F,\tau)} \le 2. \quad (20)$$

5) Signal-to-noise probability ratio (SNPR) is denoted by AUC_{SNPR} and defined by

$$0 \le \text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}}. \quad (21)$$

The above detection measures will be used to evaluate the quantitative performance of a detector. Specifically, they can be categorized according to their functionalities.

1) *TD (Target Enhancement):* $\text{AUC}_{(D,\tau)}$ and AUC_{TD}.
2) *BS (Contrast Enhancement):* $\text{AUC}_{(F,\tau)}$, AUC_{BS}, $\text{AUC}_{(D,\tau)}$, and AUC_{SNPR}.
3) *Detector's Effectiveness:* $\text{AUC}_{(D,F)}$, AUC_{TDBS}, and AUC_{ODP}.

For those who are interested in these measures, we refer the details to [46].

B. HYDICE Image to Be Studied

An airborne hyperspectral digital imagery collection experiment (HYDICE) scene shown in Fig. 6(a) was used to analyze comparative performance among various target detectors and anomaly detectors. It was collected in August 1995 from a flight altitude of 10 000 ft. There are 15 square panels in Fig. 6(a) with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m. Due to the ground sampling distance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

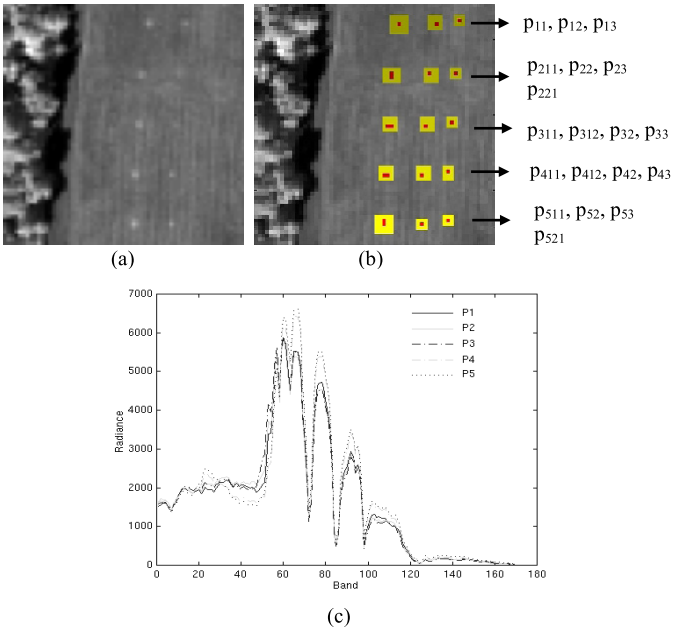


Fig. 6. (a) HYDICE panel scene that contains 15 panels. (b) Ground-truth map of spatial locations of 19 R panel pixels. (c) Spectra of $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$.

approximately 1.56 m, each of the panels in the first column except that the first row contains two panel pixels highlighted by red, p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 1. All the remaining 11 panels in Fig. 6 contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} and p_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 6(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. Fig. 6(c) plots the five-panel spectral signatures, $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$ obtained from Fig. 6(b), where the ith panel signature denoted by $\mathbf{p}_i$ was generated by averaging the red panel center pixels in row i. These five panel spectral signatures were used as desired target signature for CEM.

The best advantage of using this image scene is the ground truth provided in Fig. 6(b) about the spatial locations of each of 19 R panel pixels shown in Fig. 6(b) so that the exact spatial position of each panel pixel can be located to be used to evaluate each detector in the detection of 19 R panel pixels up to pixel level.

The VD of this scene 18 is used in [40]. Thus, in the following experiments, $n_{\text{BSam}} = 2\text{VD}^{\text{NP}}_{\text{HFC}}(\text{P}_\text{F} \le \beta = 10^{-3}) = 18$ was used as the number of bands to be sampled. It should be noted that $n_{\text{BSam}} = \text{VD}^{\text{NP}}_{\text{HFC}}(\text{P}_\text{F} \le \beta = 10^{-3}) = 9$ is exactly the bandwidth B used by the Nyquist rate discussed in Section VII, i.e., $n_{\text{BSam}} = \text{VD}^{\text{NP}}_{\text{HFC}}(\text{P}_\text{F} \le \beta = 10^{-3}) = B$.

C. Hyperspectral Target Detection by CEM

Fig. 7(a)–(h) shows the histograms of three 2-D AUC values, $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, and $\text{AUC}_{(\text{F},\tau)}$, and their derived five detection measures, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, and AUC_{ODP} by (18)–(22) that were calculated from CEM-detection maps of panel pixels in five different rows using the four versions of BSam, RUBSam, EUBSam, IRBSam, and IFBSam, and six BS methods, BF-MinV-BP, BF-MaxV-BP, BF-SF-CTBS, BF-SB-CTB, CBS, and UBS, for comparison. The results obtained by full bands are also included in Fig. 7.

Three intriguing findings can be observed from the results in Fig. 7(a)–(h).

1) To evaluate target detection performance, BSel performed better than BSam in terms of $\text{AUC}_{(\text{D},\tau)}$ in Fig. 7(b) and AUC_{TD} in Fig. 7(d).
2) On the other hand, to evaluate BKG suppression, BSam outperformed BSel and full bands in the sense of $\text{AUC}_{(\text{F},\tau)}$ in Fig. 7(c), AUC_{BS} in Fig. 7(e), and AUC_{SNPR} in Fig. 7(g).
3) However, to evaluate over detection performance, using full bands apparently performed better than both BSam and BSel in terms of $\text{AUC}_{(\text{D,F})}$ in Fig. 7(a), AUC_{TDBS} in Fig. 7(f), and AUC_{ODP} in Fig. 7(h). This made sense since using full-band information can uncover many unknown interferers and subtle substances that cause many false alarms. According to the detection theory, more false alarms also imply better TD.

Nevertheless, on the whole, IRBSam performed the best in general among all the BSam and BSel methods. All the four BSam versions also performed reasonably well and their results were comparable to but slightly worse than the results obtained by full bands.

The experimental results in Fig. 7(a)–(h) revealed that three common misconceptions occurred in target detection. One is that to evaluate the effectiveness of a detector, we need to look into three measures, TD, BS, and ODP. This leads to a second misconception. Solely relying on one single detection metric is not reliable and sometime may be misleading. Finally, a third one is that the traditional 2-D ROC curve, $\text{AUC}_{(\text{D,F})}$ in Fig. 7(a), widely used in hyperspectral target detection cannot be used to evaluate the performance of a detector in BS and overall detection performance. For example, according to Fig. 7(a), all the test methods produced very close $\text{AUC}_{(\text{D,F})}$ values within 10^{-4} error threshold. With such tiny differences, it is extremely difficult to conclude that one is better than another. This phenomenon has been reported in many published works in the literature. The 3-D ROC curve-derived detection measures developed in Section IX-A indeed provide an effective means of evaluating the performance of a detector in all aspects, TD, BS, and ODP.

Similar to Fig. 7, Fig. 8 also shows the histograms of $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, $\text{AUC}_{(\text{F},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, and AUC_{ODP} calculated from H-CEM-detection maps of panel pixels in five different rows using the four versions of BSam, RUBSam, EUBSam, IRBSam, and IFBSam, and six BS methods, BF-MinV-BP, BF-MaxV-BP, BF-SF-CTBS, BF-SB-CTB, CBS, and UBS, where the results obtained by full bands are also included for comparison. Since H-CEM was shown in [46] to perform better than CEM, its better results were also reflected in Fig. 8. Nevertheless, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

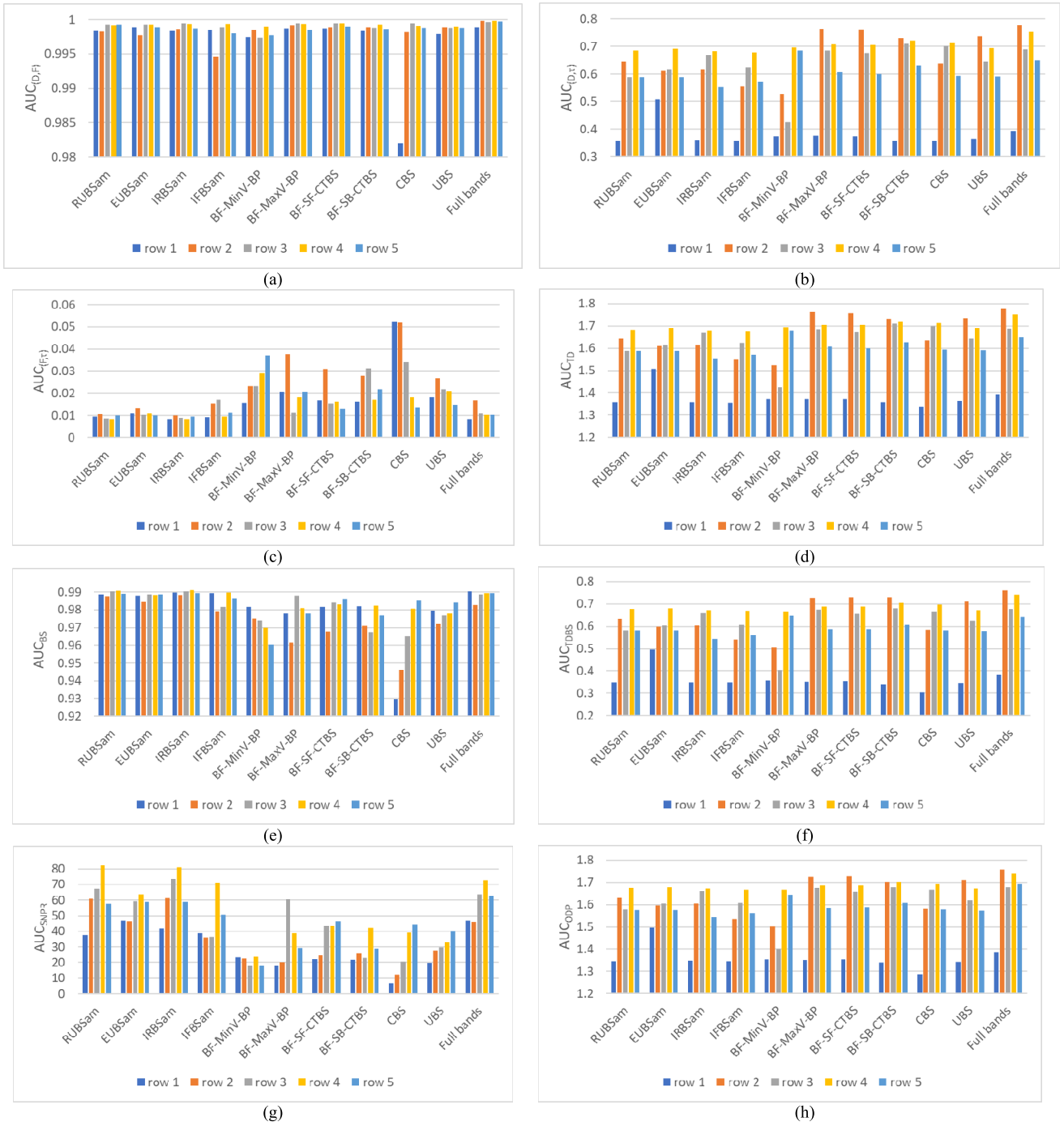


Fig. 7. Histograms of AUC values produced by CEM using different BSam and BSel methods for detecting panel pixels in all five rows. (a) $AUC_{(D,F)}$. (b) $AUC_{(D,\tau)}$. (c) $AUC_{(F,\tau)}$. (d) AUC_{TD}. (e) AUC_{BS}. (f) AUC_{TDBS}. (g) AUC_{SNPR}. (h) AUC_{ODP}.

same conclusions drawn from Fig. 7 were also applied to Fig. 8 where IRBSam performed the best in general among all the BSam and BSel methods. In particular, the performance of TD and BS between using full bands and BSam was more evidential for H-CEM than CEM. The results clearly showed that using full-band information could do better TD than BSam. On the other hand, BSam actually did better than using full bands. However, overall speaking, all the four BSam versions performed reasonably well and their results were comparable to but slightly worse than the results obtained by full bands.

D. Hyperspectral Anomaly Detection

Anomaly detection is not target detection because there is no prior knowledge about the targets it is looking for [48]. Thus, CEM is not applicable to anomaly detection. A commonly used anomaly detector is the one developed by the RX detector developed in [49], referred to as K-AD, which uses the sample covariance matrix $\mathbf{K}$, and R-AD developed in [50], which uses sample correlation matrix $\mathbf{R}$ and CRD [51]. Fig. 9 shows the histograms of $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, and AUC_{ODP} calculated from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.



Fig. 8. Histograms of AUC values produced by H-CEM using different BSam and BSel methods for detecting panel pixels in all five rows.

the anomaly detection maps of panel pixels in five different rows by K-AD, R-AD, and CRD using the four versions of BSam, RUBSam, EUBSam, IRBSam, and IFBSam, along with two BSel methods, CBSel and UBSel, and full bands for comparison. It should be noted that Fig. 9 does not include the results of BF-MinV-BP, BF-MaxV-BP, BF-SF-CTBS, and BF-SB-CTB because they are target-constrained BSel methods.

As shown in Fig. 9, several findings are interesting and different from target detection. For the anomaly detection performance of $AUC_{(D,F)}$, $AUC_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, and AUC_{SNPR}, using four versions of BSam performed significantly better than using BSel methods and outperformed full bands, while using BSel methods and full bands performed better in $AUC_{(D,\tau)}$, AUC_{TD}, AUC_{TDBS}, and AUC_{ODP}. However, it should be noted that there is a crucial difference between target detection and anomaly detection. In target detection, the targets to be detected are assumed to be known. In this case, TD is more crucial than BS. On the contrary, anomaly detection does not have prior target knowledge. Thus, it must rely on BS. In this case, BS is more critical than TD in anomaly detection. This explained why CRD performed better than K-AD and R-AD ever though K-AD and R-AD performed significantly better than CRD in TD and ODP.

In this case, all the four BSam versions performed reasonably well and their results were comparable to but slightly worse than the results obtained by full bands.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.



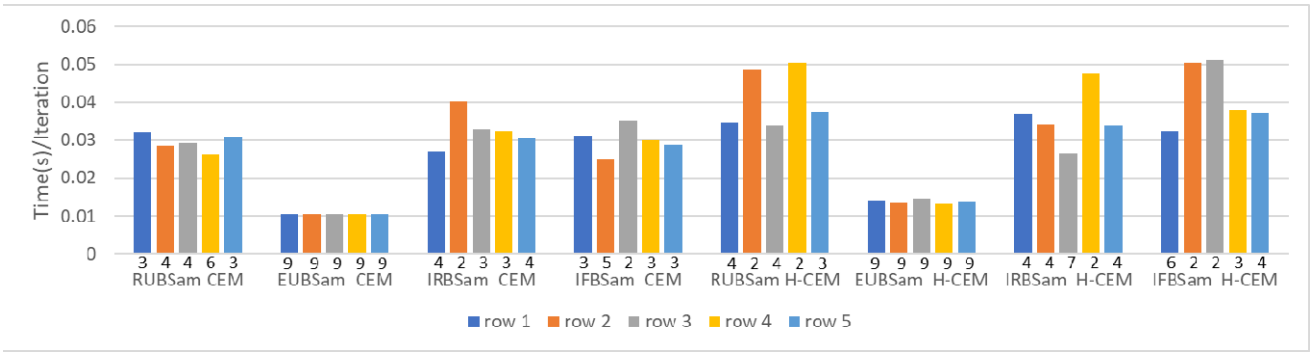
Fig. 9. Histograms of AUC values produced by K-AD, R-AD, and CRD-DW-STO using different BSam and BSel methods.

Two observations are being mentioned. One is that K-AD and R-AD using UBSel could do better than their using full bands, but this was not true for CRD. Another is somewhat surprising to see that CRD could not perform universally better than K-AD and R-AD regardless of using BSel and full bands. Generally speaking, compared to K-AD and R-AD, CRD did better in BS but worse in TD. Because there is no prior knowledge available for anomaly detection, BS is generally more crucial than TD. As a result, CRD was shown to perform better than K-AD and R-AD in many reports. However, using eight different detection measures, Fig. 9 shows different stories.

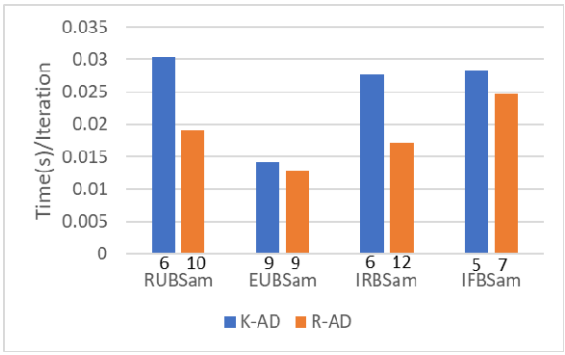
E. Time Complexity Analysis

Since BSam is not BSel, comparing BSam to BSel is sort of like comparing apples to oranges. Therefore, the time complexity analysis is conducted among the four versions of BSam methods. Furthermore, BSam involves an iterative process to reduce the uncertainty caused by BSam. Thus, the total number of iterations required to complete an iterative process varies with each BSam. Thus, to make a fair comparison, we calculated the running time per iteration. Fig. 10(a) and (b) shows the computing times (seconds)/per iteration by running CEM and H-CEM as target detectors and K-AD, R-AD, and CRD as

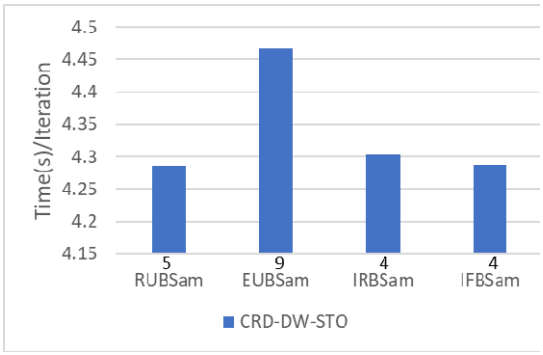
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)



(c)

Fig. 10. Computing times required by CEM, H-CEM, K/R-AD, and CRD using BSam, RUBSam, EUBSam, IRBSam, and IFBSam, per iteration. (a) Target detection by CEM and HCEM. (b) Anomaly detection by K-AD and R-AD. (c) Anomaly detection by CRD.

anomaly detectors using BSam, RUBSam, EUBSam, IRBSam, and IFBSam, respectively, where the total number of iterations for a particular BSam method is listed right underneath its bar histogram for reference. The computer environment used for experiments was 2.90 GHz Intel Corei7 CPU and 32 GB of memory.

As shown in Fig. 10(a), RBSam generally required more running time than UBSam did with EUBSam requiring the least running time. This is because the running time/iteration of EUBS is exactly the same as one time running CEM. Since H-CEM implemented a hierarchy of CEMs, its running time was greater than CEM as expected but not too much by only 1.5–2 times. As for anomaly detection, Fig. 10(b) and (c) shows a significant difference in running time between K-AD, R-AD, and CRD with two different tables where CRD required more running time in order of 10^2 in Fig. 10(c) than K-AD and R-AD did in Fig. 10(b). This is because CRD used dual windows to slide pixel by pixel to perform anomaly detection. This process is very time-consuming.

X. Applications to Hyperspectral Classification

In this section, we present another application of BSam in hyperspectral image classification where the processor was specified by one of the most recently developed classifiers, EPF-based classifiers [52], and their improved iterative counterparts, IEPF classifiers [53]. This is similar to target detection in Section IX where CEM and its improved H-CEM were used as target detectors for performance evaluation.

Although there are many image scenes available on the website [54] that can be used for HSIC, two real hyperspectral images are of particular interest. One is the Purdue Indiana Indian Pines that has an imbalanced class issue and the other is University of Pavia that has a very complex background (BKG). Both images were used to conduct a comprehensive study and analysis.

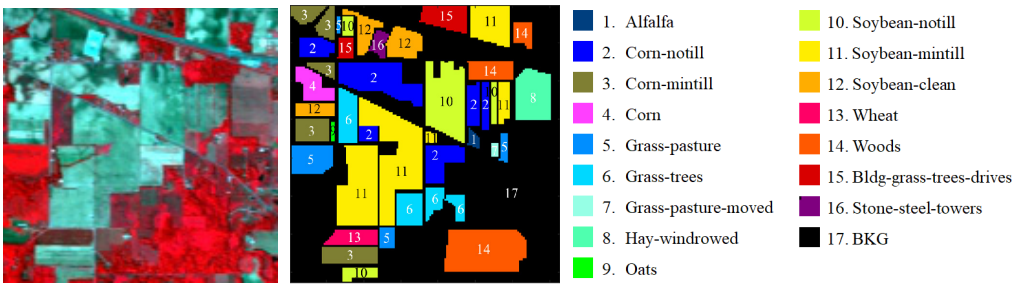
A. Purdue Indiana Indian Pines

The first image used for experiment is Purdue University's Indiana Indian Pines shown in Fig. 11(a). It is a well-known Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) image scene with the ground truth of 16 class maps plus BKG in Fig. 11(b) and class labels in Fig. 11(c). It was recorded in June 1992 using 200 bands with 20 water absorption bands (bands 104–108, 150–163, and 220) removed. Table I tabulates all the 16 classes of interest along with the number of data samples in each class where four small classes, classes 1, 7, 9, and 16, contain less than 100 data samples and three large classes, classes 2, 11, and 14, contain more than 1000 data samples. These seven classes can be used to demonstrate the issue of imbalanced classes in classification performance. Thus, when a training sample size is small, how to select individual size of training samples for each class, namely, n_m^{training} for class C_m, is challenging. This issue has been investigated in [55] and [56]. Since the work [55] is available on website, its method was used to select the training sample size for each of the classes listed in Table II with a total number of 500 trainings samples being selected for experiments.

B. University of Pavia

A second hyperspectral image used for experiments is shown in Fig. 12(a), which is different from AVIRIS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 11. Purdue Indiana Indian Pines. (a) False-color composite image. (b) Reference map. (c) Classes by colors.

TABLE I

CLASS LABELS OF INDIAN PINES WITH NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (46)	Alfalfa	class 7 (28)	Grass/pasture-mowed	class 13 (205)	Wheat
class 2 (1428)	Corn-notill	class 8 (478)	Hay-windrowed	class 14 (1265)	Woods
class 3 (830)	Corn-mintill	class 9 (20)	Oats	class 15 (386)	Bldg-grass green-drives
class 4 (237)	Corn	class 10 (972)	Soybeans-notill	class 16 (93)	Stone-steel towers
class 5 (483)	Grass/pasture	class 11 (2455)	Soybeans-mintill	class 17 (10249)	BKG
class 6 (730)	Grass/trees	class 12 (593)	Soybeans-clean		

TABLE II

NUMBER OF TRAINING SAMPLES AND TEST SAMPLES ALLOCATED INTO EACH CLASS FOR PURDUE INDIANA PINES

Class	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**Total**
n_{training}	23	36	35	32	32	35	14	32	10	36	43	34	33	40	34	31	500
n_{test}	23	1391	792	203	448	696	14	444	10	937	2416	560	172	1227	354	62	9749

TABLE III

CLASS LABELS OF UNIVERSITY OF PAVIA WITH NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (6631)	Asphalt	class 5 (1345)	Painted metal sheets	Class 9 (947)	Shadows
class 2 (18649)	Meadows	class 6 (5029)	Bare soil	Class 10 (164624)	BKG
class 3 (2099)	Gravel	class 7 (1330)	Bitumen		
class 4 (3064)	Trees	class 8 (3682)	Self-blocking bricks		

scenes, the Purdue data considered in Section V. It was collected by the ROSIS-03 satellite sensor over an urban area surrounding the University of Pavia, Italy. It is of size $610 \times 340 \times 115$ with a spatial resolution of 1.3 m/pixel and a spectral coverage ranging from 0.43 to 0.86 μm with spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Nine classes of interest are considered for this image. Fig. 12(b) shows its ground-truth map along with color class labels in Fig. 12(c). According to [57], the precision rate (PR) from this dataset was very low less than 30% twice lower than that obtained from the Purdue dataset, indicated that this scene has more complicated BKG than the other. Table III also tabulates the number of data samples in parentheses collected for each class where most classes are relatively very large except only one class, class 9 with $947 < 1000$ data samples. Thus, there is no imbalanced class issue. In this case, a total number of 500 training samples were also selected with 55 or 56 training samples selected from each class listed in Table IV.

C. Classification Measures to Be Used for Experiments

The following several classification measures defined in [57] will be used for experiments. Let n_{mm} be the number of signal samples in the mth class correctly classified into the mth class $\tilde{C}_m$, n_{jm} be the number of data samples in the mth class C_m but actually classified into the jth class $\tilde{C}_j$, M be the number of classes, C_m be the set of data samples in the mth class by ground truth, $n_m = \sum_{j=1}^{M} n_{jm}$ be the number of data samples in C_m, and N be the total number of data samples, $N = \sum_{m=1}^{M} n_m$.

In traditional HSIC, the popular performance measurements are class accuracy (A) and overall accuracy (OA) that are given by

$$P_A(C_m) = \text{accuracy of the } m\text{th class} = \frac{n_{mm}}{\sum_{j=1}^{M} n_{jm}} = \frac{n_{mm}}{n_m} \tag{22}$$

$$P_{\text{OA}} = \frac{1}{N}\sum_{m=1}^{M} n_{mm} = \sum_{m=1}^{M} \frac{n_m}{N} P_A(C_m) \tag{23}$$

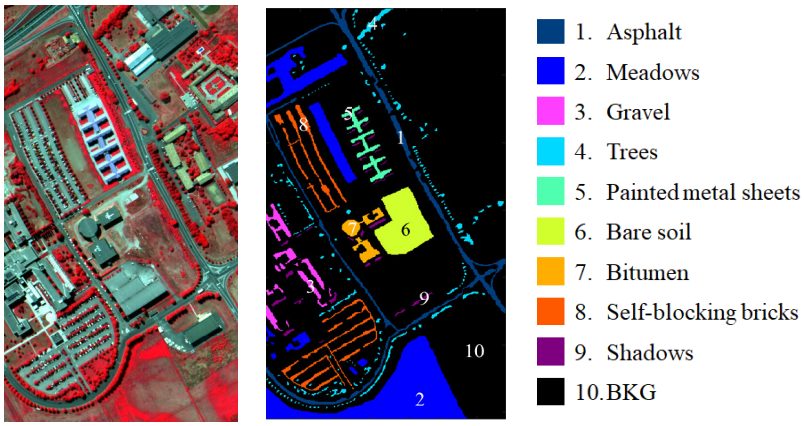
which shows that P_{OA} utilizes the sample ratios as weights to average the accuracy of each class. In addition, we can consider all classes to be equally likely by letting $(n_m/N) = (1/M)$ to further define P_{AA} as average accuracy (AA) by the number of classes, M, expressed as

$$P_{\text{AA}} = \frac{1}{M}\sum_{m=1}^{M} \frac{n_{mm}}{n_m} = \frac{1}{M}\sum_{m=1}^{M} P_A(C_m). \tag{24}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

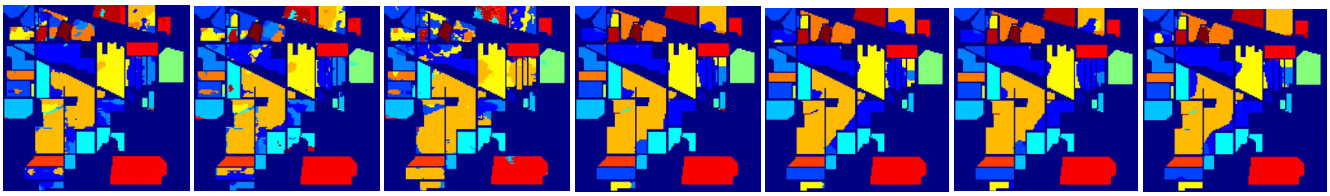
TABLE IV
NUMBER OF TRAINING SAMPLES AND TEST SAMPLES ALLOCATED INTO EACH CLASS FOR UNIVERSITY OF PAVIA DATA

Class	1	2	3	4	5	6	7	8	9	Total
n_{training}	56	56	55	56	56	55	56	55	55	500
n_{test}	6575	18593	2044	3008	1289	4974	1274	3627	892	42276



(a) (b) (c)

Fig. 12. University of Pavia. (a) False-color composite image. (b) Reference map. (c) Classes by colors.



(a) (b) (c) (d) (e) (f) (g)

Fig. 13. Classification results of Indian Pines dataset using EPF with (a) full bands. Two BS methods: (b) UBS and (c) CBS. Four BSam methods: (e) RUBSam, EUBSam (f) IRBSam (g) IFBSam.

Another measure is PR given by

$$P_{\text{PR}}(C_m) = \text{precision rate of } C_m = \frac{\tilde{n}_{mm}}{\tilde{n}_m} \tag{25}$$

where $\tilde{n}_m = \sum_{j=1}^{M} \tilde{n}_{mj}$ is the total number of data samples that are to be classified into the mth class and $\tilde{n}_{mj}$ is the number of data samples classified into the mth class but supposed to be in the jth class. Thus, the average PR (APR), P_{APR}, in correspondence to P_{AA} in (24) is defined by

$$P_{\text{APR}}(\text{BKG}) = \frac{1}{M}\sum_{m=1}^{M} P_{\text{PR}}(\tilde{C}_m). \tag{26}$$

Since n_{mm} is identical to $\tilde{n}_{mm}$, $\tilde{N} = \sum_{i=1}^{M} \tilde{n}_i$ is equal to $N = \sum_{i=1}^{M} n_i$. As a result, an overall PR (OPR), P_{OPR}, can also be defined as the counterpart of P_{OA} in (23) by

$$\begin{aligned} P_{\text{OPR}}(\text{BKG}) &= \sum_{i=1}^{M} \left(\frac{\tilde{n}_i}{\tilde{N}}\right) P_{\text{PR}}(\tilde{C}_i) \\ &= \sum_{i=1}^{M} \left(\frac{n_i}{N}\right) P_A(C_i) = P_{\text{OA}} \end{aligned} \tag{27}$$

which implies that P_{OPR} is essentially the same P_{OA}, provided that BKG class samples are not included for classification.

D. Experimental Results and Discussion

1) Purdue Data: According to [16], VD estimated for the Purdue data was $n_{\text{BSam}} = \text{VD}^{\text{NP}}_{\text{NWHFC}}(\text{P}_\text{F} \le \beta = 10^{-4}) = 18$, which was used as n_{BSam} in (16). Fig. 13 shows the classification maps produced for the Purdue data by EPF using full bands, two BSel methods, UBSel and CBSel, in [3] and the four versions of BSam, RUBSam, EUBSam, IRBSam, and IFBSam. It should also be noted that CBSel and USel were implemented due to the reason that like BSam, they are also independent of applications. Table V also tabulates their values of P_A, $P_{\text{OA}}P_{\text{AA}}$, $P_{\text{APR}}(\text{BKG})$, and $P_{\text{OPR}}(\text{BKG})$ calculated by (23)–(28) from Fig. 13 along with their iteration numbers and running time where the best results are boldfaced. In particular, $P_{\text{APR}}(\text{BKG})$ calculated by (27) and $P_{\text{OPR}}(\text{BKG})$ calculated by (28) are included to see the effect of BKG on BSel and BSam methods. As shown by the results in Fig. 13 and Table V, when the training sample size is small with only 500 training samples, EPF performed poorly using full bands and two BSel methods, UBS and CSB, with P_{OA} from 83.35% down to 66.88%, P_{AA} from 89.94% down to 67.79%, and P_{APR} from 85.98% down to 66.56%. Surprisingly, when BSam was used, the three accuracy rates were significantly improved by more than 10% with P_{OA} up to 93.35%–94.39%, P_{AA} up to 96.43%–96.75%, and P_{APR} up to 94.04%–95.30%. Interestingly, all the four versions of BSam performed very closely. This indicated that BSam was very robust and stable. Since the IEPF developed in [53] was shown to be superior to most existing spectral–spatial classifiers, the same experiments were also conducted for IEPF. Fig. 14 shows the classification maps produced for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

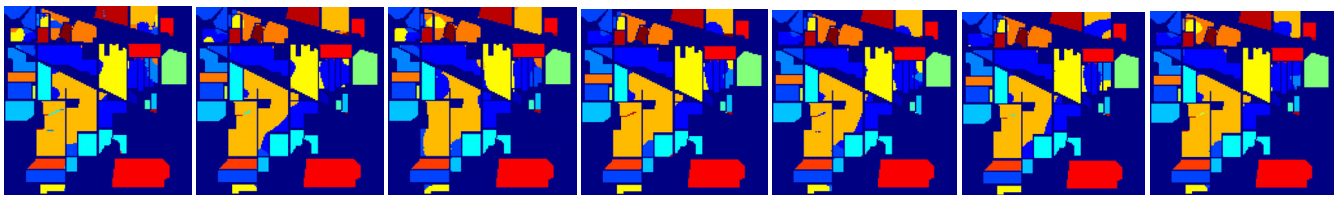


Fig. 14. Classification results of Indian Pines dataset using IEPF with (a) full bands. Two BSel methods: (b) UBS (c) CBS. Four BSam methods: (d) RUBSam, (e) EUBSam, (f) IRBSam, and (g) IFBSam.

TABLE V

AVERAGED P_A, P_{OA}, P_{AA}, AND P_{APR} RESULTING FROM TEN TIMES EXPERIMENT BY EPF USING FULL BANDS, TWO BSEL METHODS, UBS AND CBS, AND FOUR BSAM METHODS, RUBSAM, EUBSAM, IRBSAM, AND IFBSAM, FOR PURDUE DATA WHERE THE NUMBER OF SELECTED BANDS IS 18

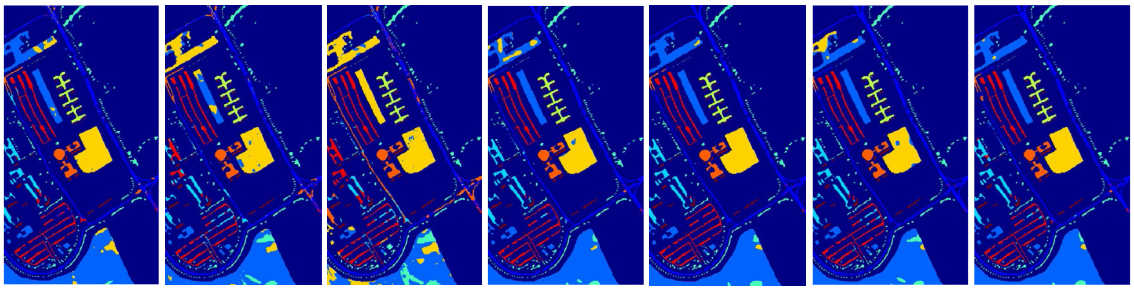
method	Full bands EPF	UBSel-EPF	CBSel-EPF	RUBSam-EPF	EUBSam-EPF	IRBSam-EPF	IFBSam-EPF
n_{training}	500	500	500	500	500	500	500
Class	$\mathrm{P_A}(C_m)$	$\mathrm{P_A}(C_m)$	$\mathrm{P_A}(C_m)$	$\mathrm{P_A}(C_m)$	$\mathrm{P_A}(C_m)$	$\mathrm{P_A}(C_m)$	$\mathrm{P_A}(C_m)$
1	**100** (0)	98.05 (1.9)	95.65 (4.59)	99.57 (1.38)	**100** (0)	99.57 (1.38)	99.13 (1.83)
2	72.44 (5.85)	57.64 (13.21)	44.6 (7.43)	89.78 (3.67)	89.76 (3.56)	**90.88** (2.44)	88.23 (4.97)
3	71.26 (8.93)	56.34 (4.32)	49.16 (17.37)	97.92 (1.35)	97.56 (1.49)	**98.15** (1.25)	96.21 (2.83)
4	99.62 (0.81)	100 (0)	92.78 (10.15)	99.95 (0.15)	**100** (0)	**100** (0)	99.90 (0.31)
5	93.91 (2.30)	70.83 (25.69)	57.77 (31.7)	**97.89** (2.73)	97.27 (3.64)	**97.84** (2.77)	97.33 (3.23)
6	97.45 (2.97)	91.94 (4.84)	95.97 (2.75)	99.13 (1.08)	98.85 (1.30)	**99.27** (0.87)	98.95 (1.07)
7	**96.43** (0)	96.79 (3.13)	52.86 (35.23)	91.54 (7.65)	92.31 (7.25)	93.08 (5.67)	95.39 (3.97)
8	99.94 (0.20)	99.31 (1.33)	99.50 (0.95)	99.98 (0.07)	**100** (0)	99.98 (0.07)	**100** (0)
9	95.00 (11.6)	23 (30.29)	0 (0)	100 (0)	**100** (0)	**100** (0)	**100** (0)
10	81.52 (6.59)	64.66 (7.72)	66.78 (5.92)	92.39 (3.27)	**93.47** (3.11)	93.28 (4.07)	93.42 (3.79)
11	77.15 (6.05)	72.34 (6.2)	65.75 (12.1)	**89.00** (4.11)	87.17 (3.66)	87.94 (5.51)	86.54 (3.60)
12	76.48 (9.37)	35.95 (9.59)	25.23 (12.9)	**96.89** (2.78)	96.10 (2.28)	96.25 (2.61)	95.89 (2.84)
13	99.41 (0.21)	99.46 (0.15)	99.51 (0)	99.31 (0.65)	**99.43** (0.38)	99.37 (0.42)	99.31 (0.53)
14	93.15 (6.08)	96.71 (2.41)	94.21 (6.20)	99.09 (1.01)	**98.88** (0.92)	98.69 (1.33)	98.72 (1.20)
15	86.09 (7.01)	55.21 (14.06)	45.02 (16.8)	**99.66** (1.08)	99.18 (2.31)	98.89 (2.44)	99.06 (2.18)
16	99.14 (2.72)	99.03 (3.06)	**99.78** (0.46)	93.87 (4.8)	92.58 (6.19)	94.84 (3.63)	94.84 (4.67)
$\mathrm{P_{OA}}$	83.35 (2.16)	72.62 (3.76)	66.88 (3.41)	**94.39** (1.14)	93.87 (1.21)	94.29 (1.18)	93.36 (1.15)
$\mathrm{P_{AA}}$	89.94 (1.18)	76.08 (3.89)	67.79 (3.97)	96.62 (0.78)	96.41 (0.82)	**96.75** (0.44)	96.43 (0.63)
$\mathrm{P_{APR}}$	85.98 (1.63)	74.93 (4.60)	66.56 (3.95)	95.30 (1.67)	94.62 (2.00)	**95.20** (1.35)	94.04 (1.68)
$\mathrm{Po_{PR}}$(BKG)	40.63 (1.05)	35.40 (1.83)	32.60 (1.66)	**43.77** (0.53)	43.53 (0.56)	43.72 (0.55)	43.29 (0.53)
$\mathrm{P_{APR}}$(BKG)	47.49 (3.01)	37.85 (3.31)	32.70 (3.48)	44.60 (1.74)	**44.63** (1.47)	44.33 (1.86)	44.01 (1.29)
Iteration	NA	NA	NA	11.8 (2.74)	12	11.8 (1.93)	7.60 (0.84)
Time (s)	2.67 (0.95)	0.43 (0.36)	0.50 (0.42)	8.28	8.60 (0.24)	8.63 (2.05)	4.56 (0.90)

Purdue data by IEPF using full bands, UBSel and CBSel, and the four versions of BSam, RUBSam, EUBSam, IRBSam, and IFBSam. Table VI also tabulates their values of P_A, P_{OA}, P_{AA}, and P_{APR} calculated from Fig. 14 along with their iteration numbers and running time where the best results are boldfaced. All the results in Tables V and VI were run ten times with standard deviation included in parentheses. As shown by the results in Fig. 14 and Table VI, IEPF significantly improved EPF nearly 10% in all three accuracy rates. Most striking finding was that IEPF also tremendously improved the two BSel methods nearly 20%–25% also in all three accuracy rates. With such great improvement, BSam still managed to improve all three accuracy rates by 1.5%–2%. Once again, all the four versions of BSam performed very closely. This indicated that BSam was very robust and stable. However, by looking at the values of P_{PR}(BKG) and P_{APR}(BKG) that calculate PR including BKG data samples via (19) and (20), it is clear that BKG had a significant impact on BSel and BSam methods that could not improve PR like what they did in OA and AA but rather reduce its PR performance. This demonstrated an important fact that full-band information is crucial to reduce the BKG effect on classification.

2) University of Pavia: According to [16], $n_{\mathrm{BSam}} = \mathrm{VD}^{\mathrm{NP}}_{\mathrm{NWHFC}}(\mathrm{P_F} \leq \beta = 10^{-4})$ is 14. Fig. 15 shows the classification maps produced for the University of Pavia by EPF using full bands, UBSel and CBSel, and the four versions of BSam, RUBSam, EUBSam, IRBSam, and IFBSam. Table VII also tabulates their values of P_A, P_{OA}, P_{AA}, P_{APR}, P_{PR}(BKG), and P_{APR}(BKG) calculated from Fig. 15 along with their iteration numbers and running time where the best results are boldfaced. Specifically, P_{PR}(BKG) and P_{APR}(BKG) are included to assess the effect of BKG on BSel and BSam methods.

Comparing the results in Fig. 15 with Fig. 13 and Table V with Table VII, one immediate finding was that the EPF performance was significantly better for the University of Pavia than for the Purdue data, while UBSel and CBSel still performed poorly. This is mainly due to the fact that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.



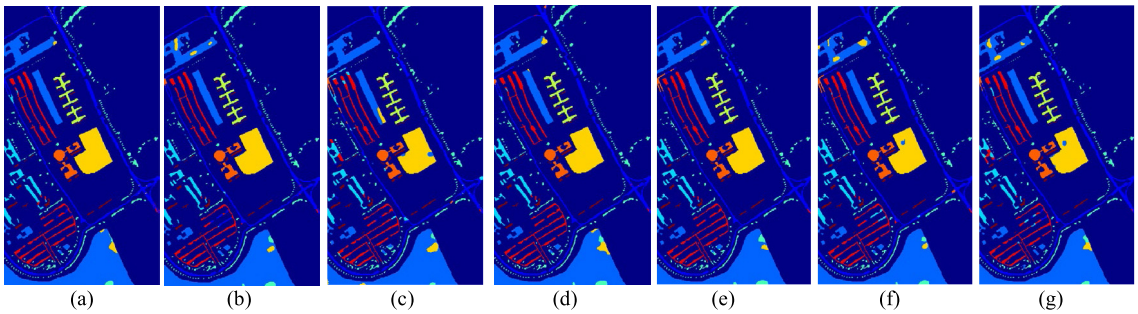
(a) (b) (c) (d) (e) (f) (g)

Fig. 15. Classification results of University of Pavia dataset using IEPF with (a) full bands. Two BSel methods: (b) UBS (c) CBS. Four BSam methods: (d) RUBSam, (e) EUBSam, (f) IRBSam, and (g) IFBSam.

TABLE VI

AVERAGED P_A, P_{OA}, P_{AA}, AND P_{APR} RESULTING FROM TEN TIMES EXPERIMENT BY IEPF USING FULL BANDS, TWO BSEL METHODS, UBS AND CBS, AND FOUR BSAM METHODS, RUBSAM, EUBSAM, IRBSAM, AND IFBSAM, FOR PURDUE DATA WHERE THE NUMBER OF SELECTED BANDS IS 18

method	Full bands IEPF	UBSel-IEPF	CBSel-IEPF	RUBSam-IEPF	EUBSam-IEPF	IRBSam-IEPF	IFBSam-IEPF
n_{training}	500	500	500	500	500	500	500
Class	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$
1	97.83 (2.29)	**100** (0)	94.78 (5.34)	**100** (0)	99.13 (1.83)	99.57 (1.38)	99.13 (1.83)
2	87.54 (3.68)	88.92 (4.5)	86.27 (4.25)	89.74 (3.45)	90.70 (3.71)	90.28 (4.17)	86.27 (7.24)
3	94.43 (3.17)	96.35 (3.2)	96.33 (2.41)	97.27 (2.86)	96.79 (2.74)	96.16 (4.03)	**97.28** (2.55)
4	99.90 (0.31)	**100** (0)	99.8 (0.42)	**100** (0)	**100** (0)	**100** (0)	99.90 (0.21)
5	97.09 (4.10)	97.13 (3.17)	96.53 (2.43)	**97.76** (2.76)	97.69 (3.04)	97.40 (2.92)	96.94 (3.05)
6	99.11 (0.74)	99.23 (0.86)	99.17 (1.32)	99.11 (0.99)	99.01 (1.05)	98.97 (1.10)	**99.33** (1.02)
7	94.62 (3.71)	93.85 (4.86)	93.08 (8.47)	93.08 (6.74)	91.54 (7.65)	94.62 (5.19)	**95.39** (3.97)
8	**100** (0)	99.98 (0.07)	**100** (0)	99.98 (0.07)	99.95 (0.10)	**100** (0)	**100** (0)
9	**100** (0)	**100** (0)	**100** (0)	**100** (0)	**100** (0)	**100** (0)	**100** (0)
10	88.73 (6.74)	**93.40** (2.88)	89.97 (3.86)	92.44 (3.56)	93.04 (4.08)	92.03 (3.40)	91.79 (4.26)
11	86.13 (4.47)	86.89 (4.89)	86.54 (4.65)	**90.01** (4.57)	88.42 (4.26)	88.46 (5.70)	88.92 (3.65)
12	95.27 (5.23)	95.37 (3.10)	93.52 (4.29)	97.03 (1.67)	96.57 (2.06)	**97.26** (1.87)	96.18 (2.76)
13	98.97 (0.59)	99.26 (0.55)	**99.71** (0.56)	99.25 (0.47)	99.43 (0.38)	99.43 (0.38)	99.37 (0.42)
14	97.62 (1.76)	98.44 (1.52)	98.30 (1.46)	**99.53** (0.48)	99.01 (1.43)	99.09 (1.16)	98.81 (1.28)
15	**99.69** (0.51)	99.09 (2.77)	97.73 (2.77)	99.15 (1.93)	98.67 (2.98)	**99.66** (0.88)	99.26 (1.54)
16	**95.81** (5.49)	92.90 (5.39)	91.78 (4.59)	93.55 (5.48)	92.42 (4.37)	**94.36** (4.32)	93.39 (4.65)
P_{OA}	92.41 (1.39)	93.49 (1.38)	92.48 (1.57)	**94.62** (1.26)	94.26 (1.46)	94.15 (1.20)	93.64 (1.20)
P_{AA}	95.79 (0.84)	96.30 (0.67)	95.22 (0.68)	**96.74** (0.71)	96.40 (1.05)	96.70 (0.72)	96.37 (0.57)
P_{APR}	93.92 (1.25)	93.96 (1.42)	92.77 (1.89)	**95.61** (1.53)	95.11 (2.16)	94.80 (2.16)	94.05 (2.81)
Po_{PR}(BKG)	42.85 (0.64)	43.38 (0.63)	42.88 (0.73)	**43.88** (0.59)	43.70 (0.67)	43.65 (0.56)	43.42 (0.56)
P_{APR}(BKG)	45.64 (1.47)	43.95 (0.15)	43.06 (2.05)	45.00 (1.56)	**44.60** (1.76)	44.05 (1.69)	43.62 (1.49)
Iteration	NA	NA	NA	7.1 (3.72)	12	7.8 (2.39)	3.20 (0.42)
Time (s)	23.32 (3.21)	1.02 (1.25)	6.90 (1.38)	10.75 (5.54)	17.74 (2.36)	11.88 (3.51)	5.41 1.12)



(a) (b) (c) (d) (e) (f) (g)

Fig. 16. Classification results of University of Pavia dataset using IEPF with (a) full bands. Two BSel methods: (b) UBS (c) CBS. Four BSam methods: (d) RUBSam, (e) EUBSam, (f) IRBSam, and (g) IFBSam.

University of Pavia has very complicated BKG than the Purdue data. In this case, it requires full-band information to perform classification well, as shown in Table VII. However, EPF using the four BSam methods still did better than EPF using full bands. Interestingly, if we performed the same experiments by simply replacing EPF with IEPF, their classification results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

AVERAGED P_A, P_{OA}, P_{AA}, AND P_{APR} RESULTING FROM TEN TIMES EXPERIMENT BY EPF USING FULL BANDS, TWO BSEL METHODS, UBS AND CBS, AND FOUR BSAM METHODS, RUBSAM, EUBSAM, IRBSAM, AND IFBSAM, FOR UNIVERSITY OF PAVIA DATA WHERE THE NUMBER OF SELECTED BANDS IS 14

method	Full bands EPF	UBSel-EPF	CBSel-EPF	RUBSam-EPF	EUBSam-EPF	IRBSam-EPF	IFBSam-EPF
n_{training}	500	500	500	500	500	500	500
Class	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$
1	92.44 (2.54)	81.46 (2.78)	60.42 (5.20)	96.77 (1.89)	96.63 (1.55)	**97.35** (1.76)	96.45 (1.72)
2	94.08 (2.94)	73.42 (4.75)	53.72 (14.36)	93.79 (2.71)	93.64 (3.59)	**94.80** (2.44)	93.67 (3.70)
3	87.84 (7.07)	69.30 (18.9)	26.83 (15.32)	96.77 (2.93)	**97.70** (2.78)	95.40 (3.18)	95.47 (4.77)
4	91.51 (1.59)	91.80 (1.71)	93.47 (1.45)	97.13 (1.55)	**97.43** (0.90)	97.25 (0.95)	96.86 (1.53)
5	99.98 (0.05)	**99.99** (0.03)	99.89 (0.12)	99.96 (0.07)	99.91 (0.08)	99.95 (0.07)	99.94 (0.08)
6	**97.69** (2.81)	86.22 (8.26)	89.59 (7.37)	96.43 (2.72)	97.09 (2.17)	95.76 (4.75)	95.91 (2.94)
7	99.26 (0.75)	99.21 (0.55)	**99.98** (0.07)	99.85 (0.42)	99.32 (1.05)	99.19 (1.33)	99.38 (0.96)
8	**97.32** (1.17)	90.50 (7.83)	86.35 (9.97)	96.48 (2.16)	96.93 (1.89)	96.78 (2.07)	96.78 (1.94)
9	**100** (0)	**100** (0)	**100** (0)	99.03 (0.62)	98.78 (0.77)	98.73 (0.69)	98.99 (0.75)
P_{OA}	94.51 (0.97)	80.98 (1.79)	67.23 (5.85)	95.66 (1.12)	95.73 (1.41)	**96.06** (1.09)	95.42 (1.78)
P_{AA}	95.57 (0.81)	87.99 (1.32)	78.92 (1.68)	97.36 (0.41)	**97.49** (0.58)	97.25 (0.64)	97.05 (0.77)
P_{APR}	93.77 (1.09)	81.85 (2.04)	72.43 (2.45)	94.99 (0.85)	95.06 (1.25)	**95.54** (1.07)	94.89 (1.37)
P_{OPR}(BKG)	19.49 (0.20)	16.70 (0.37)	13.87 (1.21)	19.50 (0.23)	19.51 (0.29)	**19.58** (0.22)	19.45 (0.36)
P_{APR}(BKG)	**22.71** (1.43)	19.10 (1.04)	15.58 (1.40)	21.20 (0.92)	21.89 (1.22)	21.29 (1.07)	21.29 (1.20)
Iteration	NA	NA	NA	6.6 (2.21)	7	7.4 (1.17)	6.7 (1.15)
Time (s)	6.95 (0.49)	2.16 (0.08)	2.53 (0.07)	12.58 (5.49)	12.06 (0.62)	14.97 (2.83)	12.76 (2.77)

TABLE VIII

AVERAGED P_A, P_{OA}, P_{AA}, AND P_{APR} RESULTING FROM TEN TIMES EXPERIMENT BY IEPF USING FULL BANDS, TWO BSEL METHODS, UBS AND CBS, AND FOUR BSAM METHODS, RUBSAM, EUBSAM, IRBSAM, AND IFBSAM, FOR UNIVERSITY OF PAVIA DATA WHERE THE NUMBER OF SELECTED BANDS IS 14

method	Full bands IEPF	UBSel-IEPF	CBSel-IEPF	RUBSam-IEPF	EUBSam-IEPF	IRBSam-IEPF	IFBSam-IEPF
n_{training}	500	500	500	500	500	500	500
Class	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$
1	**97.66** (1.32)	96.97 (1.73)	93.68 (3.12)	97.18 (1.86)	96.81 (1.59)	97.28 (2.01)	96.36 (1.29)
2	96.28 (3.28)	92.15 (3.90)	**96.55** (2.01)	92.93 (2.34)	92.90 (2.68)	92.61 (3.44)	92.67 (3.67)
3	**97.81** (1.83)	96.77 (3.10)	92.18 (4.98)	96.16 (2.70)	97.65 (2.89)	96.93 (1.64)	95.27 (2.83)
4	96.93 (1.30)	97.11 (1.06)	**97.36** (1.38)	96.68 (1.40)	96.43 (1.23)	96.58 (2.1)	96.83 (1.41)
5	**99.93** (0.09)	99.90 (0.12)	99.96 (0.07)	99.88 (0.14)	99.92 (0.14)	99.9 (0.09)	99.83 (0.14)
6	98.97 (0.99)	94.13 (4.32)	**96.66** (3.06)	96.52 (3.04)	97.97 (1.71)	97.26 (2.94)	95.09 (4.10)
7	**100** (0)	99.84 (0.32)	99.18 (1.23)	99.95 (0.08)	99.55 (1.00)	99.97 (0.07)	99.72 (0.64)
8	**97.63** (2.31)	97.32 (1.48)	91.97 (3.37)	97.45 (1.48)	97.21 (1.70)	97.26 (1.62)	96.29 (1.97)
9	99.92 (0.12)	98.56 (0.77)	99.93 (0.12)	98.77 (1.13)	98.00 (0.84)	**99.23** (0.92)	98.65 (1.05)
P_{OA}	**97.35** (1.33)	94.76 (1.65)	95.82 (1.10)	95.37 (1.12)	95.48 (1.08)	95.36 (1.22)	94.82 (1.59)
P_{AA}	**98.35** (0.22)	96.97 (0.64)	96.39 (0.80)	97.28 (0.77)	97.38 (0.48)	97.44 (0.41)	96.75 (0.58)
P_{APR}	**96.75** (1.41)	94.76 (1.14)	94.19 (1.32)	94.81 (1.28)	94.50 (0.77)	94.88 (1.02)	94.32 (1.16)
P_{OPR}(BKG)	**19.84** (0.27)	19.31 (0.34)	19.53 (0.22)	19.44 (0.23)	19.46 (0.22)	19.43 (0.25)	19.33 (0.32)
P_{APR}(BKG)	**22.91** (1.31)	21.39 (1.30)	21.40 (1.28)	20.99 (1.31)	21.47 (1.34)	21.04 (1.31)	20.63 (0.98)
Iteration	NA	NA	NA	3.2 (0.78)	7	3.2 (0.42)	2.8 (0.42)
Time (s)	30.91 (14.02)	16.12 (3.02)	18.44 (3.19)	19.33 (4.44)	30.01 (2.75)	19.84 (1.99)	21.43 (5.67)

in Fig. 16 and Table VIII showed that IEPF indeed produced the best results and the BSam methods could not outperform IEPF using full bands as it did for EPF in Fig. 15 and Table VII as well as for the Purdue data in Figs. 13 and 14 and Tables V and VI. In Table VIII, the results obtained by full bands are highlighted red, which were better than the boldfaced results obtained by BSam and BSel methods. Table VIII also shows that all BSel and BSam methods performed very similarly, but CBSel seemed to have a slightly better classification performance than the BSam methods. It should also be noted that all the results in Tables VII and VIII were run ten times with standard deviation included in parentheses.

As a summary, the experiments conducted for the Purdue and University of Pavia datasets demonstrated some interesting findings that have been overlooked in hyperspectral image classification. When a dataset such as Purdue has imbalanced classes, classification using BSel methods and BSam methods can effectively improve the full-band classification. On the other hand, if a dataset such as University of Pavia has a complicated background, then P_{PR}(BKG) calculated by (26) and P_{APR}(BKG) calculated by (27) will be very low, for example, only 19% and 22% using full bands for the University of Pavia compared to the Purdue data that have P_{PR}(BKG) and P_{APR}(BKG) with 45%–47%. Thus, the classification using BSel methods and BSam methods cannot

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

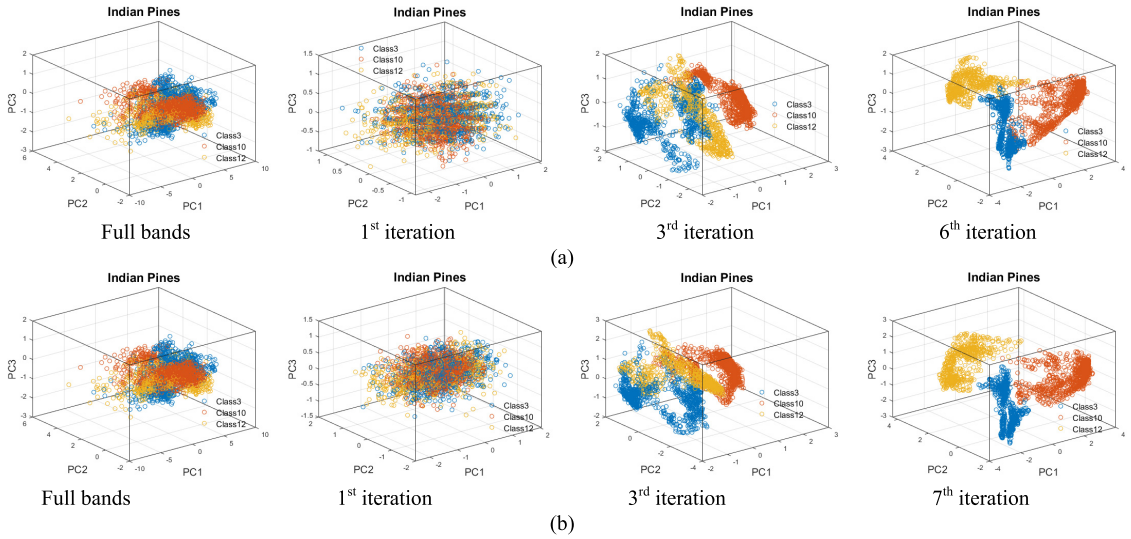


Fig. 17. 3-D representation using the first three PCs of classes 3, 10, and 12 for Indian Pines by (a) IFBSam-EPF-G-g and (b) IRBSam-EPF-G-g.

improve PR rates for the University of Pavia, but they could for the Purdue data because full bands provide crucial information to cope with BKG.

XI. Illustrative Examples to Validate the Use of SFs

As noted in Section I, BSam is a band process without including spatial information. The idea of the ITS developed in [25] enables BSam to include its much needed spatial information to improve its performance. A set of SFs is used to capture spatial information among processed data samples. It then added such additional spatial-filtered information via iterative feedback loops to expand the current data cubes so as to increase the reliability of the selected training samples. In the meantime, it also reduced the uncertainty caused by randomly selected training samples so as to improve the system performance. ITS provides a feasible solution to BSam in terms of addressing data sparsity issue. In other words, through the iterative process, data samples are more and more being compacted and clustered in low-dimensional representations corresponding to data sparsity with only a few nonzero components. In other words, this new data cube will be more compacted and more robust to random data/BSam. In terms of CS terminology, the new data cube will be more sparser than the original data cube. If the same process is repeated over and over again iteratively by feeding the spatial information captured by SFs back and added to the currently being processed data, then new expanded data cubes will keep growing through the entire iterative process. As expected, new data structures become more denser, compact, robust, and less random. Eventually, when the iterative process is terminated, the final dataset will be well-structured. This is the key idea of IBSam.

In order to validate the significant use of SFs for BSam, the SFs used to conduct classification experiments in Section X-C were once again used to demonstrate how scattering classes can be sparsely compacted in a very low-dimensional data space via iteratively feeding the spatial-filtered classification maps back to further reduce uncertainty for misclassification errors.

To have a better visual display, Fig. 17(a) and (b) shows only 3-D representations of the first three principal components (PCs) of three classes, classes 3, 10, and 12 for the Purdue data in Fig. 11 produced with different increasing numbers of iterations by IFTS-EPF-G-g and IRTS-EPF-G-g using full bands, respectively. The reason that classes 3, 10, and 12 were selected was because their classification accuracy rates were low (71.26%, 81.52%, and 76.48% for classes 3, 10, and 12, respectively) in Table V due to their data samples badly mingled with each other. As shown in Fig. 17(a) and (b), the data samples in these three classes were still mingled heavily after one iteration. Interestingly, as the number of iterations is increased by the spatial-filtered information feedback process, the three classes of data samples began to be gradually separated well. When the iterative process was terminated, these three classes are separated by IFBSam-EPF-G-g reasonably well in Fig. 17(a), but IRBSam-EPF-G-g did much better than IFBSam-EPF-G-g in Fig. 17(b) with these three classes being well-separated. This simple example suggested that we could use as few as three PCs to classify these three classes well without using full bands. This also provides the rationale and basis of BSam. If we consider that each class can be accommodated by one single spectral band, we need at least 16 bands to separate 16 classes. Table V shows that with $n_{\text{BSam}} = 18$, the OA of classes 3, 10, and 12 produced by EPF using full bands was 71.26%, 81.52%, and 76.48%, respectively, and could be significantly improved by IRBSam-EPF to 98.15%, 93.28%, and 96.25% and by IFBSam-EPF to 96.21%, 93.42%, and 95.89% for classes 3, 10, and 12 respectively. Similarly, Fig. 18(a) and (b) shows the 3-D representations of the first three PCs of three classes, classes 2, 3, and 8 for the University of Pavia in Fig. 12 produced with different numbers of iterations by IFBSam-EPF-G-g and IRBSam-EPF-G-g using full bands, respectively. As a consequence using SFs by IBSam, the final data cube can be sparsely compacted in a very low-dimensional data space as shown in the final

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

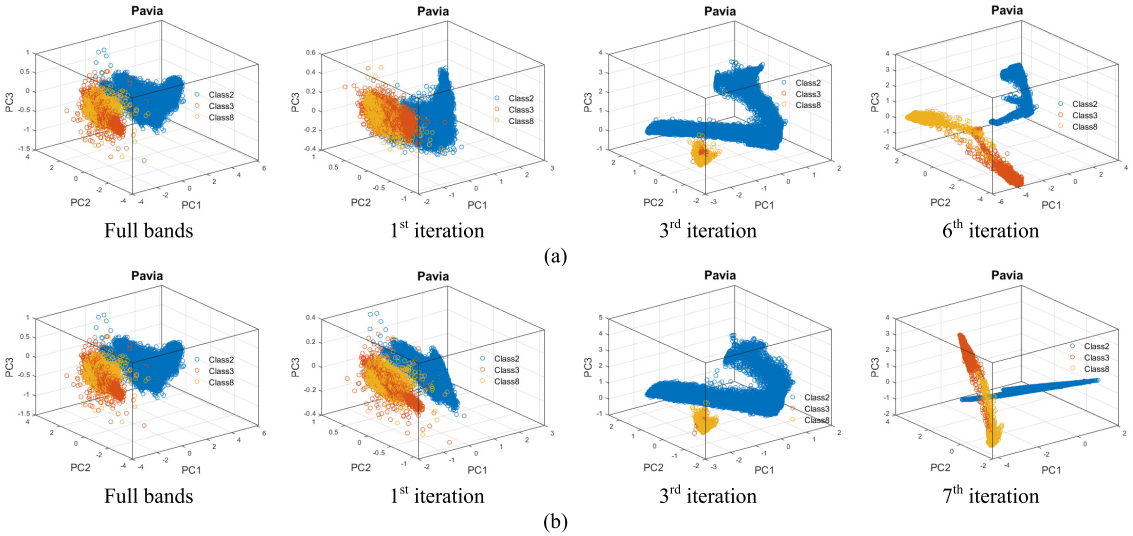


Fig. 18. 3-D representation using the first three PCs of classes 2, 3, and 8 for the University of Pavia by (a) IFBSam-EPF-G-g and (b) IRBSam-EPF-G-g.

iterations in Figs. 17 and 18 where BSam can work very effectively to cluster three different classes into three separate groups compared to those produced by full bands and the first iterations. Most importantly, despite the fact that the randomly sampled bands were not reproduceable, they only served as an intermediate passage and did not affect the final end classification results. In other words, even BSam produces different sets of sparse clusters, and the final classifications based on the resulting sparse clustered data samples were very close. This is the beauty of IBSam, which exactly corresponds to the random sensing matrix $\mathbf{\Phi}$ in (5) used by CS that is generated by a Gaussian or Bernoulli random matrix. It is also similar to a convolution neural network (CNN), which also uses SFs in its convolution layers to capture spatial information in a feedforward manner. However, its generated adaptive weights between two interlayers are in fact nonreproduceable. Nonetheless, such weights have very little impact on the final end classification results.

XII. Conclusion

This article presents a new notion of BSam with its mathematical insights from signal sampling and CS perspectives. For BSam to be effective, similar concepts of sparsity and inherence are also developed. To maximize incoherence, two approaches are derived. One is UBSam according to information theory, which assumes that maximal entropy can be achieved by the uniform distribution. The other is RBSam, which corresponds to the random sensing matrix used in CS. To address the issue of sparsity, an iterative process of using spatula filters is introduced in BSam. To estimate BSamR, the concept of VD is used to determine the number of bands to be sampled, n_{BSam}. Interestingly, it is n_{BSam} that can be used to describe sparsity order of entire hyerspectral band space. A key difference between BSam and BSel is that bands selected by BSel are deterministic, whereas the bands sampled by RUBSam and RBSam are random. The latter is a core idea derived from CS as random signal sampling, which is warranted by CS. Nevertheless, such random sampling does not have effect on the end results. In order to realize UBsam and RBsam in practical applications, their corresponding algorithms are further developed. Specifically, there is no requirement of specific rules as BSel does. Moreover, no particular applications need to be specified to implement BSam. This is similar to CS, provided that the issues of sparsity and incoherence are appropriately addressed. To illustrate its versatility, three applications are demonstrated, hyperspectral target detection, anomaly detection, and hyperspectral image classification via extensive experiments. All experimental results showed that BSam generally outperforms BSel. Last but not least, BSam introduced in this article may offer a new direction of hyperspectral band processing as a better alternative to BSel. Hopefully, in the future, we can show that what the BSamR, i.e., n_{BSam}, is important to BSam is the same as what the Nyquist rate is important to signal sampling of the signal bandwidth B corresponding to n_{BSam}.

Acknowledgment

The author would like to thank Dr. Kenneth Yeonkong Ma, Remote Sensing Signal and Image Processing Laboratory, University of Maryland at Baltimore, for his help in carrying out all experiments conducted in this article.

References

[1] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[2] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[3] C.-I. Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.

[4] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

[5] Q. Wang, Q. Li, and X. Li, "Hyperspectral band selection via adaptive subspace partition strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4940–4950, Dec. 2019.

[6] Q. Li, Q. Wang, and X. Li, "An efficient clustering method for hyperspectral optimal band selection via shared nearest neighbor," *Remote Sens.*, vol. 11, no. 3, p. 350, Feb. 2019, doi: 10.3390/rs11030350.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

[7] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.
[8] W. Zhang, X. Li, Y. Dou, and L. Zhao, "A geometry-based band selection approach for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4318–4333, Aug. 2018.
[9] Y. Wang *et al.*, "Constrained-target band selection for multiple-target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6079–6103, Aug. 2019.
[10] Y. Xu, Q. Du, and N. H. Younan, "Particle swarm optimization-based band selection for hyperspectral target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 554–558, Apr. 2017.
[11] H. Su, B. Yong, and Q. Du, "Hyperspectral band selection using improved firefly algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 68–72, Jan. 2016.
[12] G. Zhu, Y. Huang, J. Lei, Z. Bi, and F. Xu, "Unsupervised hyperspectral band selection by dominant set extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 227–239, Jan. 2016.
[13] Y. Yuan, X. Zheng, and X. Lu, "Discovering diverse subset for unsupervised hyperspectral band selection," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 51–64, Jan. 2017.
[14] C.-I Chang, L. C. Lee, B. Xue, M. Song, and J. Chen, "Channel capacity approach to band subset selection for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 10, pp. 4630–4644, Oct. 2017.
[15] L. Wang *et al.*, "Band subset selection for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 4887–4898, Sep. 2017.
[16] C. Yu, M. Song, and C.-I. Chang, "Band subset selection for hyperspectral image classification," *Remote Sens.*, vol. 10, no. 1, p. 113, Jan. 2018, doi: 10.3390/rs10010113.
[17] Q. Wang, F. Zhang, and X. Li, "Hyperspectral band selection via optimal neighborhood reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8465–8476, Dec. 2020.
[18] R. G. Gallager, *Information Theory and Reliable Communication*. Hoboken, NJ, USA: Wiley, 1968.
[19] C.-I. Chang, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification*. Norwell, MA, USA: Kluwer, 2003.
[20] C.-I Chang, "A review of virtual dimensionality for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1285–1305, Apr. 2018.
[21] E. J. Candes and M. B. Wakin, "An introduction to compressive sampling," *IEEE Signal Process. Mag.*, vol. 25, no. 2, pp. 21–30, Mar. 2008.
[22] G. R. Baraniuk, "Compressive sensing," *IEEE Signal Process. Mag.*, vol. 24, no. 4, pp. 116–124 Jul. 2007.
[23] R. Baraniuk, M. Davenport, R. DeVore, and M. Wakin, "A simple proof of the restricted isometry property for random matrices," *Constructive Approx.*, vol. 28, no. 3, pp. 253–263, Dec. 2008.
[24] M. R. Willett, F. M. Duarte, A. M. Davenport, and G. R. Baraniuk, "Sparsity and structure in hyperspectral imaging: Sensing, reconstruction, and target detection," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 116–125, Jan. 2014.
[25] C.-I Chang, K. Y. Ma, C.-C. Liang, Y.-M. Kuo, and S. Chen, "Iterative random training sampling spectral spatial classification for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3986–4007, 2020.
[26] G. Martín, J. M. Bioucas-Dias, and A. Plaza, "HYCA: A new technique for hyperspectral compressive sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2819–2831, May 2015.
[27] M. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jan. 2011.
[28] J. M. Bioucas-Dias and M. A. Figueiredo, "Alternating direction algorithms for constrained sparse regression: Application to hyperspectral unmixing," in *Proc. Hyperspectral Image Signal Process., Evol. Remote Sens.*, 2010, pp. 1–4.
[29] Q. Du, J. M. Bioucas-Dias, and A. Plaza, "Hyperspectral band selection using a collaborative sparse model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 3054–3057.
[30] P. Zhong and R. Wang, "Learning sparse CRFs for feature selection and classification of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 12, pp. 4186–4197, Dec. 2008.
[31] D. Tuia, N. Courty, and R. Flamary, "Multiclass feature learning for hyperspectral image classification: Sparse and hierarchical solutions," *ISPRS J. Photogramm. Remote Sens.*, vol. 105, pp. 272–285, Jul. 2015.
[32] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.
[33] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification via kernel sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 217–231, Jan. 2013.
[34] C. J. Della Porta, A. A. Bekit, B. H. Lampe, and C.-I Chang, "Hyperspectral image classification via compressive sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8290–8303, Oct. 2019.
[35] A. Bekit, C.-I. Chang, B. Lampe, C. J. Della Porta, and C. C. Wu, "N-FINDER for finding endmembers in compressively sensed band domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1087–1101, Oct. 2020.
[36] B. Lampe, C.-I. Chang, A. Bekit, and C. J. Della Porta, "Restricted entropy and spectrum properties for the compressively sensed domain in hyperspectral imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5642–5652, Feb. 2020.
[37] S. Theodoridis and K. Koutroumbas, *Pattern Recognition*. New York, NY, USA: Academic, 1999, p. 366.
[38] S. S. Epp, *Discrete Mathematics With Applications*, 2nd ed. Pacific Grove, CA, USA: Brooks/Cole, 1995.
[39] H. V. Poor, *An Introduction to Detection and Estimation Theory*, 2nd. ed. New York, NY, USA: Springer-Verlag, 1994.
[40] C.-I. Chang, *Hyperspectral Data Processing: Algorithm Design and Analysis*. Hoboken, NJ, USA: Wiley, 2013.
[41] J. C. Harsanyi, W. Farrand, and C.-I. Chang, "Detection of subpixel spectral signatures in hyperspectral image sequences," in *Proc. Amer. Congr. Surveying Mapping (ACSM), Amer. Soc. Photogramm. Remote Sens. (ASPRS), Annu. Conv. Expo.*, Baltimore, MD, USA, vol. 1, 1994, pp. 236–247.
[42] J. C. Harsanyi, "Detection and classification of subpixel spectral signatures in hyperspectral image sequences," Ph.D. dissertation, Dept. Elect. Eng., Univ. Maryland, Baltimore, MD, USA, 1993.
[43] W. H. Farrand and J. C. Harsanyi, "Mapping the distribution of mine tailings in the Coeur d'Alene River Valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.
[44] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.
[45] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, 2019, doi: 10.3390/rs11111310.
[46] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sensing*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.
[47] *3D ROC Analysis Software*. Accessed: Jul. 31, 2021. [Online]. Available: https://wiki.umbc.edu/display/rssipl/10.+Download
[48] C.-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Jun. 18, 2021, doi: 10.1109/TGRS.2021.3086768.
[49] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.
[50] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.
[51] L. Wei and D. Qian, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.
[52] X. Kang, S. Li, and J. A. Benediktsson, "Spectral-spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.
[53] S. Zhong, C.-I. Chang, and Y. Zhang, "Iterative edge preserving filtering approach to hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 90–94, Jan. 2019.
[54] Accessed: Jul. 31, 2021. [Online]. Available: http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes
[55] Accessed: Jul. 31, 2021. [Online]. Available: http://xudongkang.weebly.com/
[56] S. Zhong *et al.*, "Class feature weighted hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8394–8416, Nov. 2019.
[57] C.-I. Chang, "Statistical detection theory approach to hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2057–2074, Apr. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.



Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York at Stony Brook, Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana-Champaign, Urbana, IL, USA, in 1982, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He has been with the University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, since 1987, where he is a Professor with the Department of Computer Science and Electrical Engineering. He has authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real-Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection* (Springer, 2016), and *Real-Time Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017). His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE. He received the National Research Council Senior Research Associateship Award, from 2002 to 2003, sponsored by the U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, MD, USA. He also edited two books, *Recent Advances in Hyperspectral Signal and Image Processing* (2006) and *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007) and co-edited with A. Plaza a book on *High Performance Computing in Remote Sensing* (CRC Press, 2007). He is also editing a book, *Advances in Hyperspectral Image Processing Techniques* (Wiley, 2022). He also serves as an Associate Editor for *Remote Sensing* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.